Chapter – 1

INTRODUCTION

1.1 Introduction to the Internet of Things (IoT):

The Internet of Things (IoT) has revolutionized various aspects of our daily lives, including how we secure our homes. IoT technology allows everyday devices to connect to the internet and communicate with each other, making systems smarter, more efficient, and user-friendly. One of the most significant applications of IoT is in enhancing home security, where it provides a robust, automated, and reliable solution for protecting our homes.

1.2 History of :IoT:

The concept of the Internet of Things (IoT) has a rich history that dates back several decades. The idea was first proposed in the early 1980s, but it wasn't until the late 1990s that the term "Internet of Things" was coined by Kevin Ashton. Ashton, a British technology pioneer, introduced the term in 1999 during his work at Procter & Gamble, where he was involved in optimizing supply chain management using RFID (Radio-Frequency Identification) technology.

1.2.1 Early Beginnings:

The initial idea behind IoT was to connect physical objects to the internet, allowing them to communicate with each other and with users. Early experiments involved using sensors and actuators to gather data and automate processes. For example, in the 1980s, researchers at Carnegie Mellon University developed a networked Coca-Cola vending machine that could report its inventory and the temperature of the drinks.

1.2.2 Technological Advancements:

The 1990s saw significant advancements in technology that paved the way for the development of IoT. The proliferation of the internet, the miniaturization of sensors, and the development of wireless communication technologies made it possible to connect more devices to the internet. The emergence of cloud computing in the early 2000s provided the necessary infrastructure to store and process the vast amounts of data generated by IoT devices.

1.2.3 Commercial Adoption:

The commercial adoption of IoT began to take off in the 2000s, with companies exploring the potential of connected devices in various industries. The introduction of smartphones and mobile applications further accelerated the growth of IoT, making it easier for consumers to interact with IoT devices. In 2008, the number of connected devices surpassed the global human population, marking a significant milestone in the history of IoT.

1.2.4 Modern Era:

Today, IoT has become an integral part of our lives, with applications ranging from smart homes and healthcare to industrial automation and transportation. The development of new communication protocols, such as 5G, and the increasing affordability of sensors and processors continue to drive the growth of IoT. According to recent estimates, there are over 20 billion IoT devices in use worldwide, and this number is expected to reach 75 billion by 2025.

1.3 How IOT Works:

The Internet of Things (IoT) operates through a network of interconnected devices that collect, exchange, and process data. These devices, often referred to as "smart devices," are equipped with sensors, actuators, and communication modules that enable them to interact with each other and with users. Here's a detailed look at how IoT works:

1.3.1 Sensors and Actuators:

Sensors are devices that detect and measure physical properties such as temperature, humidity, light, motion, and gas concentrations. They convert these physical properties into electrical signals that can be processed by a microcontroller. Actuators, on the other hand, are devices that can perform actions based on instructions received from a controller. Examples of actuators include motors, relays, and valves.

1.3.2 Connectivity:

Connectivity is a crucial aspect of IoT, as it allows devices to communicate with each other and with central control units. Various communication technologies are used in IoT, including Wi-Fi, Bluetooth, Zigbee, LoRa, and cellular networks. The choice of communication technology depends on factors such as range, power consumption, data rate, and application requirements.

1.3.3 Data Processing:

Once data is collected by sensors, it is sent to a microcontroller or processor for processing. Microcontrollers like the Arduino Uno, equipped with the ATmega328 chip, are commonly used in IoT applications due to their versatility and ease of use. These microcontrollers can convert analog signals from sensors into digital data, process the data, and make decisions based on predefined algorithms.

1.3.4 Cloud Computing:

Cloud computing plays a vital role in IoT by providing the infrastructure needed to store, process, and analyze large volumes of data. IoT devices send data to cloud servers, where it can be processed and analyzed in real-time. Cloud platforms also enable remote access and control of IoT devices, allowing users to monitor and manage their systems from anywhere in the world.

1.3.5 User Interface:

The user interface is the final component of an IoT system, allowing users to interact with their devices. This interface can be a mobile application, a web portal, or a physical display. Users can view real-time data, receive alerts, and send commands to their devices through the interface. For example, in a smart home security system, users can receive notifications about intrusions, view live video feeds, and control security settings remotely.

1.3.6 Example of IoT in Home Security:

In a typical IoT-based home security system, various sensors are installed throughout the house to monitor different parameters. For instance, gas leakage detectors are placed near gas cylinders, temperature sensors (like LM35) are used to detect fires, light sensors are installed in critical areas to detect unauthorized use of lights, and intruder detectors are placed at entry points.
These sensors continuously send data to the central processing unit, an Arduino Uno in this case. The Arduino processes the data, converts analog signals to digital, and makes decisions based on the programmed logic. If any anomaly is detected, such as a gas leak, a sudden rise in temperature, or unauthorized entry, the system triggers alarms and sends notifications to the homeowner's smartphone via the internet.

1.4 Need for IOT:

The need for IoT arises from the increasing demand for automation, efficiency, and real-time control in various aspects of life and industry. Here are some key reasons why IoT is essential:

1.4.1 Enhanced Security:

Traditional security systems required manual monitoring and were often limited in functionality. IoT provides real-time monitoring, automated alerts, and remote access, significantly enhancing the security of homes, businesses, and public spaces. IoT-based security systems can detect and respond to threats more quickly and accurately than traditional systems.

1.4.2 Improved Efficiency:

IoT enables automation of routine tasks, leading to increased efficiency and reduced human intervention. In industrial settings, IoT can optimize manufacturing processes, reduce downtime through predictive maintenance, and improve supply chain management. In homes, IoT devices can automate lighting, heating, and security systems, saving time and energy.

1.4.3 Real-Time Monitoring and Control:

IoT allows for continuous monitoring of devices and systems, providing real-time data that can be used to make informed decisions. For example, in healthcare, IoT devices can monitor patients' vital signs and alert medical staff to any abnormalities. In agriculture, IoT sensors can monitor soil moisture levels and weather conditions to optimize irrigation and crop management.

1.4.4 Convenience and Comfort:

IoT devices make life more convenient and comfortable by automating everyday tasks and providing remote control. Smart home devices, such as thermostats, lights, and security systems, can be controlled from a smartphone or voice assistant, making it easier for homeowners to manage their homes. IoT also enables personalized experiences, such as adjusting lighting and temperature based on individual preferences.

1.4.5 Data-Driven Insights:

IoT generates vast amounts of data that can be analyzed to gain insights and make better decisions. Businesses can use IoT data to understand customer behavior, optimize operations, and develop new products and services. In smart cities, IoT data can be used to improve traffic management, reduce energy consumption, and enhance public safety.

1.4.6 Scalability and Flexibility:

IoT systems are highly scalable and can be expanded to include additional devices and sensors as needed. This flexibility makes IoT suitable for a wide range of applications, from small-scale home automation to large-scale industrial systems. IoT platforms also support interoperability, allowing devices from different manufacturers to work together seamlessly.

1.5 Applications of IoT:

IoT technology has diverse applications across various sectors, transforming industries and improving quality of life. Here are some of the key applications of IoT:

1.5.1 Smart Homes:

IoT is revolutionizing home automation by enabling devices to communicate and be controlled remotely. Smart home applications include:

- **Security Systems:** IoT-based security systems use cameras, motion sensors, and alarms to detect and respond to intrusions. Homeowners can receive real-time alerts and view live video feeds on their smartphones.
- **Energy Management:** Smart thermostats, lighting systems, and appliances can be controlled remotely to optimize energy usage and reduce costs. These devices can learn user preferences and adjust settings automatically.
- **Home Automation:** IoT devices such as smart locks, doorbells, and voice assistants enhance convenience and security. Homeowners can control access to their homes, monitor visitors, and automate routine tasks.

1.5.2 Healthcare:

IoT is transforming healthcare by enabling remote monitoring, improving patient care, and enhancing operational efficiency. Applications include:

- **Remote Monitoring:** Wearable devices and smart sensors can monitor patients' vital signs, such as heart rate, blood pressure, and glucose levels. This data is transmitted to healthcare providers in real-time, enabling timely intervention.
- **Telemedicine:** IoT enables remote consultations and diagnosis, reducing the need for in-person visits. Patients can receive medical advice and treatment from the comfort of their homes.
- **Smart Medical Devices:** IoT-connected medical devices, such as insulin pumps and pacemakers, can be monitored and controlled remotely. These devices provide real-time data to healthcare providers, improving patient outcomes.

Chapter –2

Previous Work on IOT Home Security Systems

The integration of Internet of Things (IoT) technology into home security systems has significantly enhanced their functionality and user convenience, bringing a new era of interconnected, intelligent devices that provide comprehensive and efficient security solutions. Previous work in this area has focused on developing systems that offer real-time monitoring, remote access, and automation capabilities, leveraging the power of IoT to create more responsive and user-friendly security setups.

2.1 Early Innovations and Key Developments:

One of the most notable early innovations in IoT-based home security is the Ring Video Doorbell, which allows homeowners to see, hear, and speak to visitors via a smartphone app. This device features motion detection and cloud video storage, enabling users to monitor their front door from anywhere in the world. The success of Ring demonstrates the potential of IoT to transform traditional security devices into smart, connected solutions that provide added layers of convenience and security.

Another significant development in the IoT home security landscape is the SimpliSafe system. Launched in 2006, SimpliSafe offers a DIY installation model with wireless sensors and cameras that integrate with a central monitoring service. The system provides real-time alerts and allows homeowners to control and monitor their security setup remotely through a smartphone app. SimpliSafe's approach makes home security more accessible and customizable, catering to a wide range of users and their specific needs.

2.2 Arduino and ESP8266 in IoT Security Systems:

In the realm of DIY and hobbyist projects, the use of Arduino microcontrollers and ESP8266 WiFi modules has been particularly influential. These components have enabled the creation of sophisticated IoT home security systems on a budget, demonstrating how various sensors can be integrated into a cohesive system managed via a smartphone app.

For example, a typical IoT-based home security project might involve an Arduino Uno board connected to several sensors, including gas leakage detectors, temperature sensors, intruder alarms, and light sensors. The ESP8266 module provides the necessary WiFi connectivity, allowing the Arduino to communicate with a remote server or directly with the user's smartphone. When a sensor is triggered, such as in the case of gas leakage or an unauthorized entry, the system can send real-time alerts to the homeowner, who can then take appropriate action remotely.

2.3 Comprehensive Smart Home Security Solutions:

Modern IoT home security systems go beyond simple alert mechanisms. They integrate various smart home devices to create a more holistic security solution. For instance, systems like the ADT Pulse and Vivint Smart Home offer integration with smart locks, lights, and thermostats. These systems can automate responses to security breaches, such as turning on lights or locking doors when an intruder is detected, thereby enhancing the overall security posture of the home.

ADT Pulse, for example, provides a platform where users can monitor and control their home security setup via a single app. It integrates various devices, including cameras, motion detectors, and smart locks, and provides real-time alerts and remote access. Similarly, Vivint Smart Home offers an advanced security solution that includes AI-powered analytics, enabling features like facial recognition and behavior analysis to reduce false alarms and provide more accurate monitoring.

2.4 Real-World Applications and Benefits:

The practical applications of IoT in home security are vast and varied. One of the primary benefits is the ability to receive real-time alerts and monitor one's home from anywhere in the world. This capability is particularly valuable for frequent travelers or individuals who spend long hours away from home, providing peace of mind and enhanced security.

Moreover, the integration of IoT allows for more efficient and automated security responses. For instance, if a smart smoke detector senses a fire, it can automatically trigger alarms, unlock smart locks to facilitate evacuation, and notify emergency services. This level of automation and interconnectivity significantly improves the effectiveness of home security systems.

Chapter – 3

IOT Based Home Security System Project Explanation

3.1 Circuit Diagram:

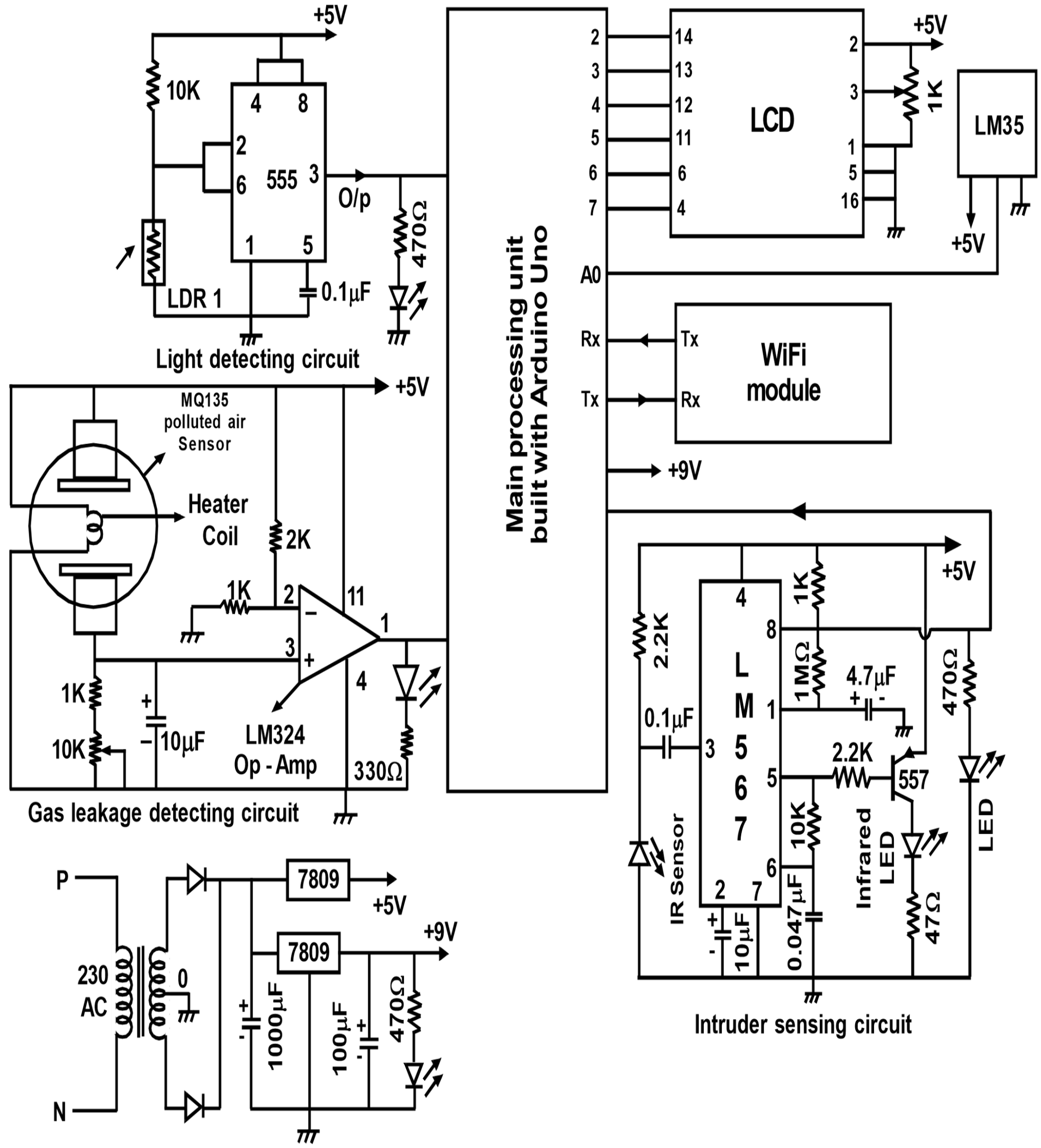

Fig 3.1 IOT based Home Security System

3.2 Project Overview:

Our project aims to develop a sophisticated home security system that can detect unauthorized intrusions, monitor environmental conditions such as temperature and light intensity, and alert homeowners to potential threats in real-time. By integrating Arduino-based microcontrollers with a range of sensors, including gas sensors, temperature sensors, light sensors, and infrared (IR) sensors, we are able to create a comprehensive security network that covers all aspects of home safety. Additionally, by incorporating Blynk IoT integration, we enable remote monitoring and control of the security system, providing homeowners with peace of mind even when they are away from home.

3.2.1 Sensor Integration:

- Gas Sensor (SMOKE): The gas sensor plays a crucial role in detecting potential gas leaks within the home environment. It continuously monitors the air quality and triggers an alarm in case of a gas leak, allowing homeowners to take immediate action to mitigate the risk.
- Temperature Sensor (LM35): The LM35 temperature sensor serves a dual purpose in our system. Firstly, it detects fire traces by monitoring any sudden spikes in temperature, indicating the presence of a fire hazard. Secondly, it provides ambient temperature readings, allowing homeowners to monitor the temperature conditions within their homes and take appropriate action if necessary.
- Light-Dependent Resistor (LDR): The LDR is used to detect changes in light intensity, distinguishing between day and night conditions. It helps conserve energy by automatically controlling lighting systems based on ambient light levels and can also alert homeowners to potential intruders by detecting sudden changes in light.
- Infrared (IR) Sensors: IR sensors are strategically placed near the main entry points of the home to detect unauthorized intrusions. They utilize infrared technology to detect body heat and movement, triggering alarms and notifying homeowners of potential security breaches.

3.2.2 Blynk IoT Integration:

In addition to sensor integration, our home security system incorporates Blynk IoT integration, allowing homeowners to monitor and control the system remotely from their smartphones or other internet-enabled devices. Blynk is a popular platform for building IoT applications, offering a user-friendly interface and a range of features for connecting hardware devices to the cloud.

3.2.3 System Architecture:

The core of our system is the Arduino Uno microcontroller, which serves as the central processing unit for interfacing with sensors and controlling system operations. The ESP8266 Wi-Fi module enables wireless connectivity, allowing data transmission to the Blynk cloud server. The Blynk mobile app provides a user-friendly interface for homeowners to monitor sensor data, receive real-time alerts, and control the security system remotely.

3.2.4 Functionality and Operation:

Upon system initialization, the Arduino Uno configures the sensor inputs and establishes communication with the ESP8266 module. The sensors continuously monitor their respective parameters, including gas levels, temperature, light intensity, and infrared activity. Any deviations from normal conditions trigger alarms and notifications, which are transmitted via the Wi-Fi module to the Blynk cloud server. Homeowners can then receive alerts on their smartphones and take appropriate action in response to security incidents.

3.2.5 User Interface:

The Blynk mobile app provides a user-friendly interface for homeowners to monitor sensor data and control the security system remotely. The app displays real-time sensor readings, system status updates, and alerts, allowing homeowners to stay informed about the security status of their homes at all times. Additionally, the app provides options for configuring alarm thresholds, setting up custom notifications, and remotely arming and disarming the security system.

Chapter – 4

Hardware Description

To prove any project work practically for the demonstration purpose, construction of the described model is essential. For this purpose suitable hardware in the form of electronic, electrical and mechanical components are essential to perform the given task. When these components are integrated together or working together, better results can be obtained from the project work. Since it is a practical oriented project work, the content presented in the abstract must be proven practically. In this regard required active hardware like IC’s and other special components must be gathered and their details must be described in this chapter to fulfill the concept of perfect project report.

Electronic hardware is Hardware, in the context of technology, refers to the physical elements that make up electronic system or electro-mechanical system, and everything else involved that is physically touchable. When an embedded system is considered, that contains a processing unit (Often microcontroller chips are preferred to build a processing unit) Sensors, control circuits that includes the motors, relays, switching devices (like power Mosfets, transistors, etc). Hardware works hand-in-hand with firmware and software to make a system function. Software is a collection of code installed into the microcontroller chip. Often LCD displays are used to monitor the system performance or results. But here in this project work LCD is not required. The function of the embedded system used here is very simple which is aimed to control the beach cleaning machine.

When a computer is considered as an example, Hardware is only one part of a computer system, but there is also firmware, which is embedded into the hardware and directly controls it. There is also software, which runs on top of the hardware and makes use of the firmware to interface with the hardware. Hardware is a surrounding term that refers to all the physical parts that make up a computer. The internal hardware devices that make up the computer and ensure that it is functional are called components, while external hardware devices that are not essential to a computer’s functions are called peripherals.

The following are the active components used in this project work.

- **Arduino processor**

- **LCD**
- **Voltage regulator**
- **LM35**
- **LDR**
- **LM567**
- **LM324**
- **IC 555**

4.1 Description of Gas leakage detector circuit:

The system designed with a gas leakage detector is quite useful for the kitchens at the domestic side and in the industries at the commercial side. Gas detector is a device which detects the presence of gas in the air in a closed room. Usually gas detectors are aimed to detect all sorts of petroleum vapors including smoke. Gas leakages are hazardous to humans, therefore as a part of a safety device to warn about gas leakages, this system is developed by which concerned persons at remote sites will be alerted. Gas detectors can be used to detect combustible, toxic (poisonous) and CO_2 gasses. Here TGS 813 is used as a gas sensor, the data sheets of this device are provided in the hardware details chapter.

TGS 813 or MQ series sensors are general-purpose sensors, which have good sensitivity characteristics to a **wide** range of gasses. This device is designed to operate at 5V-regulated supply. The most suitable application for the TGS 813 is the detection of methane, propane and butane, which makes it an excellent sensor for gas leak detectors. Whenever the sensor detects any toxic gas its output voltage will be increased and this voltage will be compared with an op-amp configured as voltage comparator. With the help of a reference voltage generated at one input of the comparator, sensor output will be compared and if it is more than the reference voltage, comparator output will become high. This high signal is fed to the Arduino processor and on receipt of this high signal this processor is programmed to display the information and the same will be transmitted through the IOT device.

TGS 813 is a general purpose Sensor that has good sensitivity characteristics to a wide range of gasses. This device is designed to operate with a stabilized 5V heater supply and a circuit voltage depends on the design. The most suitable application for the TGS 813 is the detection of domestic gas leaks.

The initial stabilization time of the TGS 8813 is very short and the relative and elapsed characteristics are very good over a long period of Operation. TGS 813 has a very low sensitivity to 'noise-gases', which considerably reduces the Problem of nuisance alarms. The TGS 8813 is most practically employed in a circuit design, which maintains circuit voltages at fixed values of 5V. This voltage rating is very practical when determining design specifications because of the wide range of available components. This makes the use of the TGS 813 an especially economical way to design low-cost, highly reliable gas detection circuits.

4.1.1 Structural specifications:

The TGS 813 is a sintered bulk semiconductor composed mainly of tio-dioxide. The semiconductor material and electrodes are deposited on a ceramic tubular former. The heater coil is located inside the ceramic former. This coil, made of 60-micron diameter wire, has a resistance of 30 ohms. The lead wires from the Sensor electrodes are a gold alloy of 80-micron diameter. The heater and lead wires are spot welded to the Sensor pins, which are arranged to fit a 6-pin miniature tube socket. The Sensor base and cover are made of nylon. The deformation temperature for this material is in excess of 240°C. The upper and lower openings in the Sensor case are covered with a flameproof double layer of 100-mesh stainless steel gauze. Independent tests confirm that this mesh will prevent a spark produced inside the flameproof cover from igniting an explosive.

4.1.2 The following is the basic test circuit:

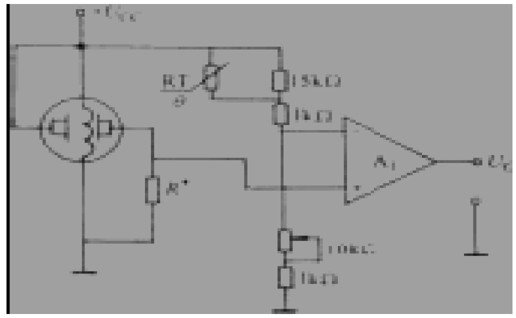

Fig 4.1.2.1 Tgs Sensor Test Circuit

The Variation in resistance of the TGS sensors measured indirectly as a Change in voltage appearing across the load resistor RL. In fresh air the current passing through the Sensor and RL in series is steady, but when a combustible gas such as propane, methane etc. comes in contact with the Sensor surface, the Sensor resistance decreases in accordance with the gas concentration present. The voltage Change across RL is the same when VC and VH are supplied from AC or DC sources. One can feel that this circuit is most suitable for evaluating the TGS 813 performance because of the ease in measuring the output signal.

The sensitivity characteristics of the TGS 813 Sensor are altered by changes in atmospheric temperature and humidity. The detection principle of the TGS is based on Chemical adsorption and desertion of gasses on the Sensor surface. Because these reactions are temperature dependent and water vapor tan is considered a gas, the effects of temperature and humidity changes cannot be eliminated from the Sensor. These effects can however be reduced by circuit design.

4.1.3 Time for initial stabilization:

A TGS Sensor, which has been stored de-energized for a long period, takes some time to reach its normal operating condition following switch on. From the moment of switch on the sensor's conductivity first rises rapidly and then falls towards its final stable value. The time taken to stabilize is a function of the sensor's storage time and atmosphere. In general, the longer the storage time, the longer the initial action time. In the case of Sensor type 8813, the initial action time reaches its maximum value after about 20 days of storage. In normal applications the initial action time will be less than 2 minutes. It is important to note that if the Sensor is placed in the air containing gas immediately after the "Initial Action", then the Rs value will appear according to the characteristics described in the next section.

Practical detector circuit using 813 Sensor

As it was mentioned above the TGS 813 is suitable for use in the detection of a wide range of gasses such as natural gas, L.P.G. and town gas. When we design a circuit employing the TGS 813 we must consider both the type and concentration of gas we wish to detect. Because of its sensitivity characteristics, the action of the Sensor will vary according to the type and concentration of gas it is detecting. Furthermore, the proper alarm Point for the detector should be determined after considering the following factors,

a. Where the Sensor is to be installed

b. Purpose of detector (gas leak, automatic fan control, air monitoring, etc.)

c. Operation of detector (Sound, light, fan control, valve control, etc.)

d. Type of gas being detected or monitored.

The output of the gas sensor is fed to the op-amp and it is configured in voltage comparator mode such that the circuit generates logic high signal when the gas sensor detects the gas leakage. The following is the description.

4.1.4 Op-AMP description:

The operational amplifier is arguably the most useful single device in analog electronic circuitry. With only a handful of external components, it can be made to perform a wide variety of analog signal processing tasks. It is also quite affordable. One key to the usefulness of these little circuits is in the engineering principle of feedback, particularly negative feedback, which constitutes the foundation of almost all automatic control processes. The op-amp used in the project work is denoted by lm324, this is a quad op-amp.

4.1.5 LM324 Quad operational amplifier:

This device consists of 14 pins. It consists of four independent, high gain, internally frequency compensated operational amplifiers which were designed specifically to operate from a single power supply over a wide range of voltages. Operation from split power supplies is also possible and the low power supply current drain is independent of the magnitude of the power supply voltage.Application areas include transducer amplifiers, dc gain blocks and all the conventional operation amplifier circuits, which now can be more easily implemented in single power supply systems. For example, the lm324 series can be directly operated off of the standard +5v power supply voltage, which is used in digital systems and will easily provide the required interface electronics without requiring the additional +15v power supplies.

The below is the figure of LM 324 quad operational device showing the function of each pin

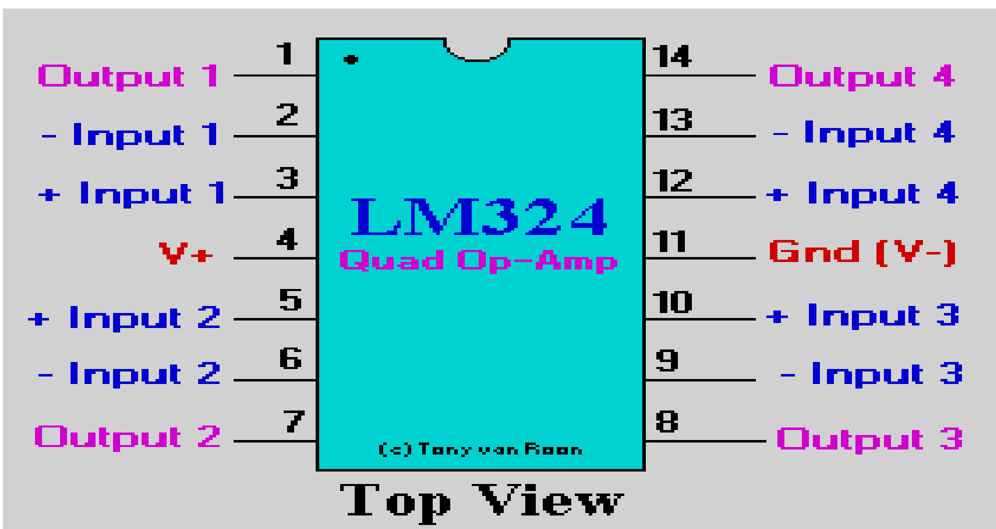

Fig 4.1.5.1 LM324 Quad operational amplifier

4.2 Description:

The LM124 series consists of four independent, high gain, internally frequency compensated operational amplifiers which were designed specifically to operate from a single power supply over a wide range of voltages. Operation from split power supplies is also possible and the low power supply current drain is independent of the magnitude of the power supply voltage.

Application areas include transducer amplifiers, DC gain blocks and all the conventional op amp circuits, which now can be more easily implemented in single power supply systems. For example, the LM124 series can be directly operated off of the standard +5V power supply voltage, which is used in digital systems and will easily provide the required interface electronics without requiring the additional ±15V power supplies.

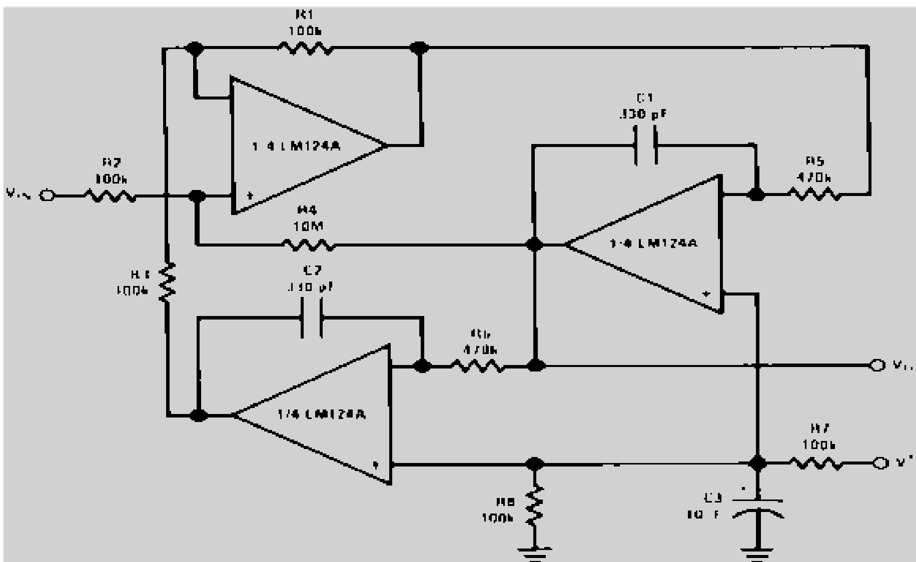

Fig 4.2.1 The LM324 series

The LM324 contains four independent high gain operational amplifiers with internal frequency compensation. The four op-amps operate over a wide voltage range from a single Power supply. Also use a split power supply. The device has low power supply current drain, regardless of the power supply voltage. The low power drain also makes the LM324 a good choice for battery operation. The LM324 series are low-cost, quad operational amplifiers with true differential inputs. They have several distinct advantages over standard operational amplifier types in single supply applications.

4.2.1 General Description:

The LM124 series consists of four independent, high gain; internally frequency compensated operational amplifiers, which were designed specifically to operate from a single power supply over a wide range of voltages. Operation from split power supplies is also possible and the low power supply current drain is independent of the magnitude of the power supply voltage.

4.2.2 Advantages:

- Eliminates need for dual supplies
- Four internally compensated op amps in a single package
- Allows directly sensing near GND and VOUT also goes to GND
- Compatible with all forms of logic
- Power drain suitable for battery operation

4.2.3 Features:

- Internally frequency compensated for unity gain
- Large DC voltage gain 100 dB
- Wide bandwidth (unity gain) 1 MHz (temperature compensated)
- Wide power supply range: Single supply 3V to 32V or dual supplies ±1.5V to ±16V
- Very low supply current drain (700 µA)-essentially independent of supply voltage
- Low input biasing current 45 nA (temperature compensated)
- Low input offset voltage 2 mV and offset current: 5 nA
- Input common-mode voltage range includes ground
- Differential input voltage range equal to the power supply voltage
- Large output voltage swing 0V to V^+ - 1.5V

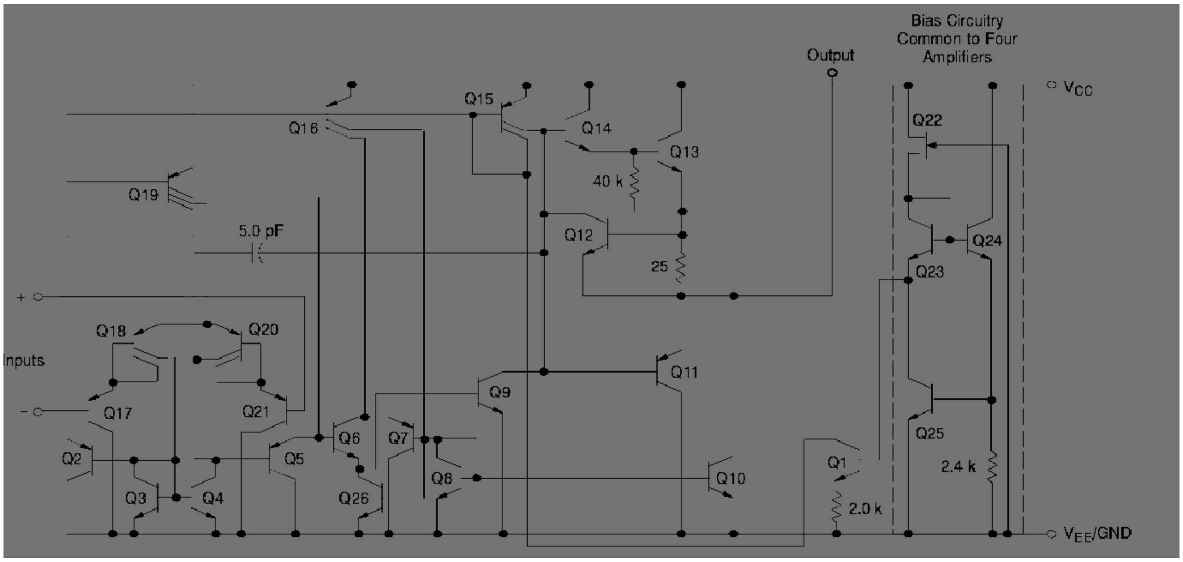

Fig 4.2.1.1 Representation Circuit Diagram (One-Fourth of Circuit Diagram)

4.2.4 Circuit Description:

The LM324 series is made using four internally compensated, two stage operational amplifiers. The first stage of each consists of differential input devices Q20 and Q18 with input buffer transistors Q21 and Q17 and the differential to single ended converter Q3 and Q4. The first stage performs not only the first stage gain function but also performs the level shifting and Tran conductance reduction functions. By reducing the Tran conductance, a smaller compensation capacitor (only 5.0 pF) can be employed, thus saving chip area. The Tran conductance reduction is accomplished by splitting the collectors of Q20 and Q18. Another feature of this input stage is that the input common mode range can include the negative supply or ground, in single supply operation, without saturating either the input devices or the differential to single ended converter. The second stage consists of a standard current source load amplifier stage.

4.3 Description of intruder sensing circuit:

The main intention of this concept is to provide a security system for the common people on the domestic side. Robbing locked houses happens to be quite common these days, because it is pretty easy for the thieves, they need not struggle much. Therefore to avoid these types of muggings, this concept is also included in this project work. The main concept involved in the system is to detect the passing person nearby the sensors, & send this interrupt information to the concerned person through IOT.

Before locking the house, the system must be activated. Initially the process begins with intruder sensors; here infrared sensors are used for detecting the presence of a person who entered into the house through the window or by breaking/tampering with the lock. Whenever the IR energy hits an object, some of the energy will be reflected, this reflected energy will be detected by the IR sensor. The same principle is used here, whenever the house is attacked by the brigands, the IR beam will be interrupted, by which a logic low signal will be generated from the trigger circuit output. Based on this signal, the Arduino Uno processor interfaced with the Wi-Fi module transmits the information.

IR sensors are used to detect the intruder and these sensors are connected to the LM 567 IC which is configured in trigger mode of operation. The following is the description.

The LM567 IC is a general-purpose tone decoder IC designed to provide a saturated transistor switch to ground when an input signal is present within the pass band. The circuit consists of two-phase detectors i.e., Q and A detectors that are driven by a voltage-controlled oscillator, which determines the center frequency of the decoder. External components are used to independently set center frequency, bandwidth and output delay.

As the IC is configured with a VCO internally, it will be generating the frequency depending on the R and C values that are connected to the 5 and 6 pins of the 567-tone decoder IC. The 5th pin is called the timing resistor (RT) and the 6th pin is called the timing capacitor (CT). As the frequency is inversely proportional to R and C values, by defining the RC network at the IC; the VCO (Voltage Controlled Oscillator) along with the Q-phase detector will be generating a particular frequency which will be coming out from the 5^{th} pin of the tone decoder IC. The I-phase detector is used to decode or compare the received frequency with the generated frequency.

The frequency generated by the tone decoder IC can be calculated using the formula:

$F = [1 / (2 \Pi R C)]$

Looking at the pin configuration in the circuit diagram of the 567-tone decoder, the 3rd pin of the IC is connected to the IR receiver (detector) and 5th pin to the IR transmitter by using a transistor. The PNP transistor SK 100 is used to drive the high efficiency IR transmitting LED with the modulating frequency generated by the IC. In simple words it can be explained like, as the signal (frequency) generated by the IC will not be having great strength, the transistor is used for amplifying the signal and the amplified signal is fed to IR transmitting LED and in order to limit the current a 470 ohms resistor is connected.

The IR has the characteristics of a laser i.e., it almost travels in a straight line with minimum expansion and like laser light it is also a monochromatic light. Another important feature of IR is that while the transmission is being done the IR transmitter and the IR receiver (detector) both should be in line of sight.

The IR receiver is connected to the 3rd pin, which is the input pin of the IC. As the IR signal is transmitted into the free space and the IR receiver detects the signal. In the free space a lot of noise signals are present and the IR signal will be received with some noise signals. So in order to eliminate the noise signals a capacitor is connected in between the IR receiver and the 3rd pin of the 567-tone decoder IC. And thus the IR received signal is fed to the IC, which will compare the received frequency signal with that of the generated frequency. The I-phase detector does this comparison and the output will be enabled when both the frequencies match, i.eThe transmitted frequency is equal to the received frequency. And if the frequencies do not match the output will not be enabled.

When both the transmitted and the received frequencies are matched, the I-phase detector enables the output pin of the 567-tone decoder i.e., the 8th pin. So when the output is enabled the output from the IC is a logic low signal. And if the frequencies do not match, the output will not be enabled and the 8th pin will logic high signal. This is due to the fact that at the 8th pin internally a transistor is present whose emitter is grounded and the collector pin the output 8th pin. And if the circuit of the 567-tone decoder is observed, we can see the supply i.e., Vcc is connected to the output pin of the IC through a resistor.

When the frequencies match, the output will be enabled by which the transistor will be conducting (ON) and the Vcc supply will be grounded through the transistor internally in the IC 567 Tone decoder itself. So a logic low signal will be received. And the same way if frequencies do not match output will not be enabled by which the transistor will not be conducting (OFF) and the supply will be coming from the output pin, which is the logic high signal.

When the IR sensors are connected facing each other, every time the output of the tone decoder IC will be a logic low signal until no obstacle comes in between the IR transmitter and the IR receiver. Unless any obstacle is present the transmitted frequency will be continuously received by the receiver and frequency will be matched with that of the generated ones and the output will be enabled. So a logic low will be the output. And if an obstacle comes in between the sensors the IR receiver will not receive the signal and the tone decoder IC checks the received frequency. As there is no received frequency the output will not be enabled, thus a logic high signal is received.

If the sensors are placed side by side, the IR transmitter will be transmitting the IR signal continuously and the receiver will receive the reflected signal when there is any obstacle. So until there is no obstacle, the receiver will not receive the IR signal, so the frequencies do not match and the output will not be enabled. So the output of the 567-tone decoder will be a logic high signal. If obstacle comes in front of the sensors, the IR signal will be reflected back which will be observed by the IR receiver and feeds the signal to the tone decoder, which checks the frequency with the generated ones and as both of them match the output will be enabled by which the internal will be in ON state and the supply will be grounded internally in the IC. So logic low signal will be received from the output of the 567-tone decoder IC.And to indicate whether the output of the sensing circuit is a logic low signal or a logic high signal, a LED is connected at the output pin of the 567-tone decoder IC. If the output is high LED will be ON state and LED will be in OFF state if the output is low.

4.4 Description of LDR based light detecting circuit:

In the field of security alarm systems, light alarm or light intensity alarm also plays a dominant role for specific applications. In general in locked houses or locked store rooms all lights will be switched off to save the electric energy. If any thief entered the house by tampering with the main door lock, he must be energizing any light source like torch light or electric light because the burglar could steal the valuables in the dark. In such a case when any light source is energized, the circuit designed with the LDR & 555 timer chip will be activated automatically and generate a logic low signal for the processing unit. Based on this signal, the main processing unit constructed with an Arduino board displays the information in the form of text message, i.e., “Light on” and the same message will be transmitted through the internet and concerned people can be monitored through smart phones.

In general, a security alarm is a system designed to detect intrusion or unauthorized entry into a building or other area such as a home or any organization. Security alarms, especially light detected devices, are used in residential and commercial properties for protection against burglary. Here in this circuit LDR (Light Dependent Resistor) is used to detect the light intensity. The following is the description of LDR.

A Light sensor is a device which is used to detect light. It has a (variable) resistance that changes with the light intensity that falls upon it. This allows them to be used in light sensing circuits. They are used in many consumer products to determine the intensity of light. For example, a light sensor is used as a part of a safety or security device like a garage door opener or a burglary alarm.

An LDR or light dependent resistor is also known as photo resistor, photocell, and photoconductor. It is a one type of resistor whose resistance varies depending on the amount of light falling on its surface. When the light falls on the surface of the LDR its resistance will be changed accordingly. These resistors are often used in many circuits where it is required to sense the presence of light. These resistors have a variety of functions and resistance. For instance, when the LDR is in darkness, then it can be used to turn ON a light or to turn OFF a light when it is in the light. A typical light dependent resistor has a resistance in the darkness of 1MOhm, and in the brightness its resistance will be decreased to less than 1 K Ohm.

This resistor works on the principle of photoconductivity. It is nothing but, when the light falls on its surface, then the material conductivity reduces and also the electrons in the valence band of the device are excited to the conduction band. These photons in the incident light must have energy greater than the band gap of the semiconductor material. This makes the electrons jump from the valence band to conduction. These devices depend on the light, when light falls on the LDR then the resistance decreases, and increases in the dark. When a LDR is kept in the dark place, its resistance is high and, when the LDR is kept in the light its resistance will decrease.

In this concept it is designed to make a light sensor circuit using LDR, 555 timer IC and a few other electronics components. This circuit detects light incident on the LDR and generates a high signal whenever the intensity of light is greater than a certain level. The resistance of LDR (Light Dependant Resistor) is inversely proportional to the intensity of light falling on it. It implies that if the intensity of incident light is high, the resistance of LDR will be less and vice versa. On the other hand, a 555 timer IC gets activated when its reset pin (Pin-4) receives a voltage greater than 0.8V. Once the IC is activated, the voltage at Pins-2&6 need to be between 1/3rd and 2/3rd of the supply voltage, for the output to be ON. For example if the voltage at the reset pin is above 0.8V and the voltage at Pins-2&6 is half the supply voltage, the output turns ON.

In the circuit, we created a voltage divider using LDR and a resistor or potentiometer. It is then connected to Pin-4 (reset) of 555 timer IC. So when it's dark, the LDR's resistance increases and so the voltage at the voltage divider drops below 0.8V, causing the 555 timer IC to turn OFF. When there is enough light, the voltage at the reset pin goes above 0.8V and the IC turns ON. Here in this circuit the reset pin is connected to +Vcc permanently by which depending on the resistance generated by the LDR according to the light intensity, the timer chip triggers at 1/3Vcc & 2/3Vcc. The 10K pot is connected in series with LDR and it is configured in a potential diving network. One end of the resistor is connected to +Vcc and one end of LDR is connected to the ground. The midpoint taken from the network is known as reference voltage and this voltage will be varied according to the light intensity that falls on the LDR. Output of this circuit is fed to the Pin no; 2 & 6 of timer IC, if the voltage is less than 1/3 Vcc, then output of the timer chip remains in zero state, similarly if the voltage is above 2/3 Vcc, timer output will become high. Likewise output status will be changed automatically.

4.5 Description of Temperature sensing circuit:

Initially, the temperature data acquired from the sensor will be converted into digital through the internal ADC of Arduino, displaying the digital temperature data in degree centigrade through LCD. In this regard, in the first step, the Arduino processor is programmed to measure and display the temperature information.

4.5.1 Temperature sensor:

The most frequently measured environmental quantity is “Temperature” This might be expected since most of the systems are affected by temperature like physical, chemical, electronic, mechanical, and biological systems. Certain chemical effects, biological processes, and even electronic circuits execute best in limited temperature ranges. Temperature is one of the most frequently calculated variables and sensing can be made either through straight contact with the heating basis or remotely, without straight contact with the basis using radiated energy in its place. There is an ample variety of temperature sensors on the market today, including Thermocouples, Resistance Temperature Detectors (RTDs), Thermistors, and Semiconductor Sensors. The sensor used here is known as Semiconductor Sensor.

Usually, a temperature sensor is a thermocouple or a resistance temperature detector (RTD) that gathers the temperature from a specific source and alters the collected information into understandable type for an apparatus or an observer. Temperature sensors are used in several applications. Here in this project work LM35 is used and it is attached to the hot body of the furnace.

LM35 is an analog, linear temperature sensor whose output voltage varies linearly with change in temperature. LM35 is three terminal linear temperature sensors from National semiconductors. It can measure temperature from-*55* degree Celsius *to* +150 degree Celsius. The voltage output of the LM35 increases 10mV per degree Celsius rise in temperature. LM35 can be operated from a 5V supply and the standby current is less than 60uA. LM35 is an integrated analog temperature sensor whose electrical output is proportional to Degree Centigrade. LM35 Sensor does not require any external calibration or trimming to provide typical accuracies. The LM35 temperature sensor can be used to detect precise centigrade temperature. The output of this sensor changes describes the linearity. The o/p voltage of this IC sensor is linearly comparative to the Celsius temperature.

LM35 is an analog temperature sensor. This means the output of LM35 is an analog signal. Microcontrollers don't accept analog signals as their input directly. We need to convert this analog output signal to digital before we can feed it to a microcontroller's input. For this purpose, here Arduino processor is used which is built in with ADC (Analog to Digital Converter). Modern day boards like Arduino and most modern day micro controllers come with inbuilt ADC.

4.5.2 Function of ADC in brief:

The Arduino processor used here has an in-built 10 bit 6 channels ADC. A-D converters are among the most widely used devices for data acquisition. Computers use binary (discrete) values, but in the physical world everything is analog (continuous). The most important parameter of an ADC is resolution. The higher resolution ADC provides a smaller step size, where step size is the smallest change that can be discerned by an ADC. In addition to resolution, conversion time is another major factor in judging an ADC. Conversion time is defined as the time it takes the ADC to convert the analog input to a digital number.

The Arduino board has six ADC channels, among those only 2 channels are used to detect the Furnace temperature & output voltage of TEG modules. The **Arduino Uno ADC** is of 10 bit resolution (so the integer values from (0-(2^10) 1023)). This means that it will map input voltages between 0 and 5 volts into integer values between 0 and 1023. So for every (5/1024= 4.9mV) per unit. So here the temperature sensor output voltage will be varied according to the temperature and its output is connected to the A0 channel. Now the Arduino is programmed to read this channel, measure the resolution and display the temperature & voltage values through LCD.

This analog to digital converter (ADC) converts a continuous analog input signal into an n-bit binary number, which is easily acceptable to a computer/microcontroller. As the input increases from zero to full scale, the output code stair steps. The full scale may be defined as 0 to 5v level. The temperature sensor used here is scaled and it raises its out voltage very linearly according to rise of temperature, meaning it generates 10 mv per degree centigrade. So if we get 1000mv (equals to 1V) from the sensor it is equivalent to 100^0 C.

4.6 Description of display section:

This chapter explains how to interface the LCD to a Microcontroller, before interfacing we have to study the operation modes of LCD's and how to program using different languages for different processors. The processor used is Arduino & following is the description of interfacing LCD with Arduino.The LCDs have a parallel interface, meaning that the microcontroller has to manipulate several interface pins at once to control the display. The interface consists of the following pins:

A **register select (RS) pin** that controls where in the LCD's memory you're writing data to. You can select either the data register, which holds what goes on the screen, or an instruction register, which is where the LCD's controller looks for instructions on what to do next.
A **Read/Write (R/W) pin** that selects reading mode or writing mode
An **Enable pin** that enables writing to the registers
8 **data pins (D0 -D7)**. The states of these pins (high or low) are the bits that you're writing to a register when you write, or the values you're reading when you read. There's also a **display contrast pin (Vo)**, **power supply pins (+5V and Gnd)** and **LED Backlight (Bolt and BOlt-)** pins that you can use to power the LCD, control the display contrast, and turn on and off the LED backlight, respectively. The process of controlling the display involves putting the data that form the image of what you want to display into the data registers, then putting instructions in the instruction register. The **Liquid Crystal Library** simplifies this for you so you don't need to know the low-level instructions.
The Hitachi-compatible LCDs can be controlled in two modes: 4-bit or 8-bit. The 4-bit mode requires seven I/O pins from the Arduino, while the 8-bit mode requires 11 pins. For displaying text on the screen, you can do most everything in 4-bit mode, so an example shows how to control a 16x2 LCD in 4-bit mode.

4.6.1 Hardware Required:

- Arduino Board
- LCD Screen (compatible with Hitachi HD44780 driver)
- pin headers to solder to the LCD display pins
- 10k ohm potentiometer
- 220 ohm resistor
- hook-up wires
- breadboard

4.6.2 Circuit:

Before wiring the LCD screen to your Arduino board we suggest soldering a pin header strip to the 14 (or 16) pin count connector of the LCD screen, as you can see in the image above. To wire your LCD screen to your board, connect the following pins:

- LCD RS pin to digital pin 12
- LCD Enable pin to digital pin 11
- LCD D4 pin to digital pin 5
- LCD D5 pin to digital pin 4
- LCD D6 pin to digital pin 3
- LCD D7 pin to digital pin 2
- LCD R/W pin to GND
- LCD VSS pin to GND
- LCD VCC pin to 5V
- LCD LED+ to 5V through a 220 ohm resistor
- LCD LED- to GND

Additionally, wire a 10k pot to +5V and GND, with its wiper (output) to LCD screens VO pin (pin3).

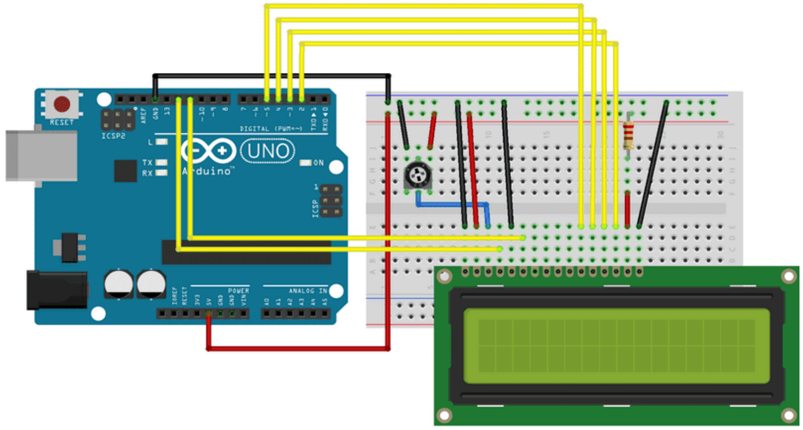

Fig 4.6.1 16*2 LCD display with Arduino Code

image developed using Fritzing. For more circuit examples, see the Fritzing project page

4.6.3 Schematic:

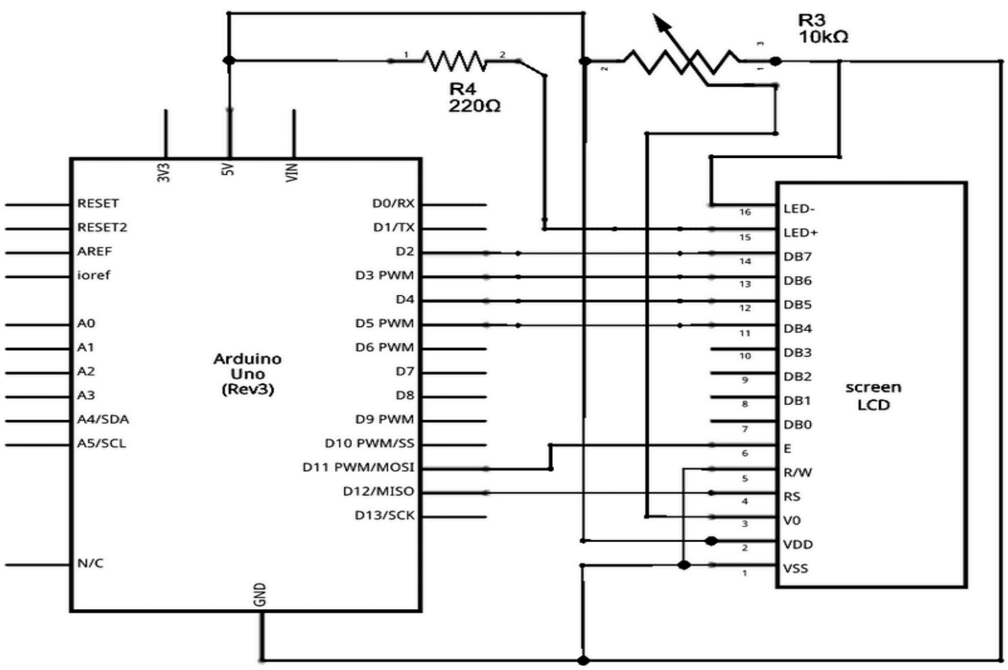

Fig 4.6.2 Displaying text on LCD Screen from the Arduino UNO

4.7 Code:

- Liquid Crystal Library - Your reference for the Liquid Crystal library.
- lcd.begin()
- lcd.print()

- lcd.setCursor()
- Blink - Control of the block-style cursor.
- Cursor - Control of the underscore-style cursor.
- Display - Quickly blank the display without losing what's on it.
- TextDirection - Control which way text flows from the cursor.
- Scroll - Scroll text left and right.
- Serial display - Accepts serial input, displays it.
- SetCursor - Set the cursor position.
- Autoscroll - Shift text right and left.

In recent years the LCD panels became very popular because of their widespread use in various electronic systems like instruments to read the parameter values, digital communications for sending or receiving the text information, data acquisition systems, etc. These display units dominate seven segment displays by providing more features to the user. The LCD system can display numbers, characters, and graphics, whereas seven segment LED's display only numbers & therefore most of the engineers prefer LCD's. The data fed to the LCD remains as it is and the same will be displayed until it gets an erase signal from the controller. The data can be stored and it can be refreshed for the next task.

The instruction command codes from the microcontroller can be sent to the LCD to clear the display, depending on the command the cursor can be brought to home position or blink the cursor. The LCD has two important resistors internally, command resistor and data register; RS pin is used to select either command register or data register. If RS = 0, the instruction command code register is selected and allows the user to send a command to clear the display. If RS = 1 the data register is selected, thereby the user is allowed to send data that is to be displayed on the LCD screen. By making the RS pin to zero, we can also check the busy flag bit to see if the LCD is ready to receive information. As already mentioned that D0 – D7 of LCD pins are 8 – bit data pins and the busy flag is D7, it can be read when R/W (Read or Write) pin is high (R/W = 0) and RS = 0, as follows; if R/W = 1, R/S = 1. When the D7 pin is high, the LCD is busy taking care of internal operations and will not accept any new information. When D7 = 0, the LCD is ready to receive new information. It is recommended to check the busy flag before writing any data to the LCD. The following table shows the list of instruction command codes.

Code	Command to LCD Instruction
1	Clear display screen
2	Return home
4	Decrement cursor (shift cursor to left)
6	Increment cursor (shift cursor to right)
5	Shift display right
7	Shift display left
8	Display off, cursor off
A	Display off, cursor on
C	Display on, cursor off
E	Display on, cursor blinking
F	Display on, cursor blinking
10	Shift cursor position to left
14	Shift cursor position to right
18	Shift the entire display to the left
IC	Shift the entire display to the right
80	Force cursor to beginning of first line
CO	Force cursor to beginning of second line
38	2 Lines and 5x7 Matrix

Table 4.6.1 Command of LCD instruction

To send any commands from the instruction command code table to the LCD, make the RS pin to zero. For data, feed a high signal to the RS pin, then send a high – to – low pulse to the E pin to enable the internal latch of the LCD. For this, the suitable program is to be prepared for LCD connections. Another suitable program is to be prepared for sending code to the LCD with checking the busy flag. Depending on the program the busy flag can be D7 of the command resistor, to read command register R/W pin must be high and RS pin must be low, and a low to high pulse for the enable pin will provide the command register. After reading the command register, if bit D7 (busy flag) is high, the LCD is busy and no information (either command or data) should be issued to it. During D7 is zero; at that time we can send data or commands to the LCD. In this method time delays are not required in the program, because we are checking the busy flag before issuing commands to the LCD. Enable line must be negative-edge triggered for the write and it should be positive-edge triggered for the read.

4.8 Description of Arduino Processor:

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are used for multiple applications in the field of electronics and communications. Arduino is a basic single board microcontroller designed to make applications, interactive controls, or environments easily adaptive. The hardware consists of a board designed around an 8-bit microcontroller, or a 32-bit ARM. Current models feature things like a USB interface, analog inputs, and GPIO pins which allow the user to attach additional boards. **GPIO** stands for General Purpose, Input, and Output. All the processors we use to have at least a few, a Raspberry Pi and an **Arduino** have a lot of General Purpose Input Output that we can design our circuits.

Introduced in 2005, the Arduino platform was designed to provide a cheaper way for hobbyists, students and professionals to create applications that play in the human interface world using sensors, actuators, motors, and other elementary products. Common applications for students or the inexperienced are simple robots or motion detectors. It offers a simple integrated IDE (integrated development environment) that runs on regular personal computers and allows users to write programs for Arduino using C or C++.

Typical prices of Arduino boards run around Rs; 300/-. Arduino boards can be purchased pre-assembled or as do-it-yourself kits. Hardware design information is available for those who would like to assemble an Arduino by hand. Over the years Arduino has been the brain of thousands of projects, from everyday objects to complex scientific instruments. A worldwide community of makers - students, hobbyists, artists, programmers, and professionals - has gathered around this open-source platform, their contributions have added up to an incredible amount of accessible knowledge that can be of great help to novices and experts alike.

Arduino was born at the Ivrea Interaction Design Institute, italy as an easy tool for fast prototyping, aimed at students without a background in electronics and programming. As soon as it reached a wider community, the Arduino board started changing to adapt to new needs and challenges, differentiating its offer from simple 8-bit boards to products for IoT applications, wearable, 3D printing, and embedded environments. All Arduino boards are completely open-source, empowering users to build them independently and eventually adapt them to their particular needs. The software, too, is open-source, and it is growing through the contributions of users worldwide.

4.8.1 Why Arduino?

Thanks to its simple and accessible user experience, Arduino has been used in thousands of different projects and applications. The Arduino software is easy-to-use for beginners, yet flexible enough for advanced users. It runs on Mac, Windows, and Linux. Teachers and students use it to build low cost scientific instruments, to prove chemistry and physics principles, or to get started with programming and robotics. Designers and architects build interactive prototypes, musicians and artists use it for installations and to experiment with new musical instruments. Makers, of course, use it to build many of the projects exhibited at the Maker Faire, for example. Arduino is a key tool to learn new things. Anyone - children, hobbyists, artists, programmers - can start tinkering just following the step by step instructions of a kit, or sharing ideas online with other members of the Arduino community.

There are many other microcontrollers and microcontroller platforms available for physical computing. Parallax Basic Stamp, Net media’s BX-24, Phidgets, MIT's Handy board, and many others offer similar functionality. All of these tools take the messy details of microcontroller programming and wrap it up in an easy-to-use package. Arduino also simplifies the process of working with microcontrollers, but it offers some advantage for teachers, students, and interested amateurs over other systems:

- Inexpensive - Arduino boards are relatively inexpensive compared to other microcontroller platforms. The least expensive version of the Arduino module can be assembled by hand, and even the pre-assembled Arduino modules cost less than $50
- Cross-platform - The Arduino Software (IDE) runs on Windows, Macintosh OSX, and Linux operating systems. Most microcontroller systems are limited to Windows.
- Simple, clear programming environment - The Arduino Software (IDE) is easy-to-use for beginners, yet flexible enough for advanced users to take advantage of as well. For teachers, it's conveniently based on the Processing programming environment, so students learning to program in that environment will be familiar with how the Arduino IDE works.
- Open source and extensible software - The Arduino software is published as open source tools, available for extension by experienced programmers. The language can be expanded through C++ libraries, and people wanting to understand the technical details can make the leap from Arduino to the AVR C programming language on which it's based. Similarly, you can add AVR-C code directly into your Arduino programs if you want to.
- Open source and extensible hardware - The plans of the Arduino boards are published under a Creative Commons license, so experienced circuit designers can make their own version of the module, extending it and improving it. Even relatively inexperienced users can build the breadboard version of the module in order to understand how it works and save money.

4.8.2 The Unique Culture of Arduino:

Some of the largest semiconductor companies have jumped into the Arduino space such as Cypress, STM, Texas Instruments, Freescale, and of course the incumbent Atmel.

What is interesting is the emergence of small cottage industries sprouting up in many unsuspected areas. This new generation of platforms is significantly different as it serves as the incubator of new designers, and a new era, as transformations of applications serve new and exciting needs.

4.8.3 Technical Description:

Pin Category	Pin Name	Details
Power	Vin, 3.3V, 5V, GND	Vin: Input voltage to Arduino when using an external power source. 5V: Regulated power supply used to power microcontroller and other components on the board. 3.3V: 3.3V supply generated by on-board voltage regulator. Maximum current draw is 50mA. GND: ground pins.
Reset	Reset	Resets the microcontroller.
Analog Pins	A0 – A5	Used to provide analog input in the range of 0-5V
Input/Output Pins	Digital Pins 0 - 13	Can be used as input or output pins.
Serial	0(Rx), 1(Tx)	Used to receive and transmit TTL serial data.
External Interrupts	2, 3	To trigger an interrupt.
PWM	3, 5, 6, 9, 11	Provides 8-bit PWM output.
SPI	10 (SS), 11 (MOSI), 12 (MISO) and 13 (SCK)	Used for SPI communication.
Inbuilt LED	13	To turn on the inbuilt LED.
TWI	A4 SDA,A5 SCA	Used for TWI communication.
AREF	AREF	To provide reference voltage for input voltage.

Table 4.8.1 Pin Description

4.8.4 Arduino Uno Technical Specifications:

Microcontroller	ATmega328P – 8 bit AVR family microcontroller
Operating Voltage	5V
Recommended Input Voltage	7-12V
Input Voltage Limits	6-20V
Analog Input Pins	6 (A0 – A5)
Digital I/O Pins	14 (Out of which 6 provide PWM output)
DC Current on I/O Pins	40 mA
DC Current on 3.3V Pin	50 mA
Flash Memory	32 KB (0.5 KB is used for Bootloader)
SRAM	2 KB
EEPROM	1 KB
Frequency (Clock Speed)	16 MHz

Table 4.8.2 Arduino Uno Technical Specifications

4.8.5 Other Arduino Boards

Arduino Nano, Arduino Pro Mini, Arduino Mega, Arduino Due, Arduino Leonardo

4.8.6 Overview

Arduino Uno is a microcontroller board based on 8-bit ATmega328P microcontroller. Along with ATmega328P, it consists of other components such as crystal oscillator, serial communication, voltage regulator, etc. to support the microcontroller. Arduino Uno has 14 digital input/output pins (out of which 6 can be used as PWM outputs), 6 analog input pins, a USB connection, A Power barrel jack, an ICSP header and a reset button.

4.8.7 How to use Arduino Board

The 14 digital input/output pins can be used as input or output pins by using pin Mode (), digital Read () and digital Write () functions in Arduino programming. Each pin operates at 5V and can provide or receive a maximum of 40mA current, and has an internal pull-up resistor of 20-50 K Ohms which are disconnected by default. Out of these 14 pins, some pins have specific functions as listed below:

- Serial Pins 0 (Rx) and 1 (Tx): Rx and Tx pins are used to receive and transmit TTL serial data. They are connected with the corresponding ATmega328P USB to TTL serial chip.
- External Interrupt Pins 2 and 3: These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value.
- PWM Pins 3, 5, 6, 9 and 11: These pins provide an 8-bit PWM output by using analogWrite() function.
- SPI Pins 10 (SS), 11 (MOSI), 12 (MISO) and 13 (SCK): These pins are used for SPI communication.
- In-built LED Pin 13: This pin is connected with an built-in LED, when pin 13 is HIGH – LED is on and when pin 13 is LOW, its off.

Along with 14 Digital pins, there are 6 analog input pins, each of which provides 10 bits of resolution, i.e. 1024 different values. They measure from 0 to 5 volts but this limit can be increased by using AREF pin with analog Reference () function.

- Analog pin 4 (SDA) and pin 5 (SCA) also used for TWI communication using Wire library.

Arduino Uno has a couple of other pins as explained below:

- AREF: Used to provide reference voltage for analog inputs with analogReference() function.
- Reset Pin: Making this pin LOW, resets the microcontroller.

4.8.8 Communication:

Arduino can be used to communicate with a computer, another Arduino board or other microcontrollers. The ATmega328P microcontroller provides UART TTL (5V) serial communication which can be done using digital pin 0 (Rx) and digital pin 1 (Tx). An ATmega16U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer. The ATmega16U2 firmware uses the standard USB COM drivers, and no external driver is needed. However, on Windows, a .inf file is required. The Arduino software includes a serial monitor which allows simple textual data to be sent to and from the Arduino board. There are two RX and TX LEDs on the Arduino board which will flash when data is being transmitted via the USB-to-serial chip and USB connection to the computer (not for serial communication on pins 0 and 1). A Software Serial library allows for serial communication on any of the Uno's digital pins. The ATmega328P also supports I2C (TWI) and SPI communication. The Arduino software includes a Wire library to simplify use of the I2C bus.

4.8.9 Arduino Uno to ATmega328 Pin Mapping

When the ATmega328 chip is used in place of Arduino Uno, or vice versa, the image below shows the pin mapping between the two.

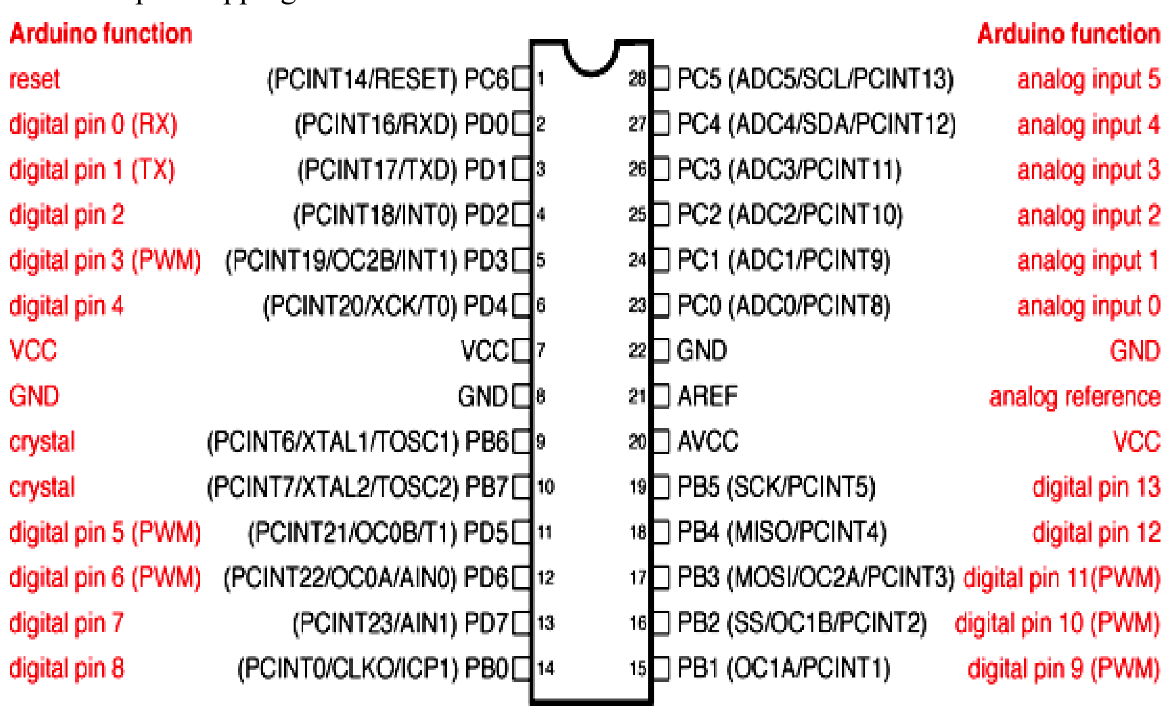

Fig 8.1 Arduino Uno to ATmega328 Pin Mapping

4.9 Description of Wi-Fi module:

The **Internet of things** (**IoT**) is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers (UIDs) and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.

The definition of the Internet of things has evolved due to the convergence of multiple technologies, real-time analytics, machine learning, sensors and embedded systems. Traditional fields of embedded systems, wireless sensor networks, control systems, automation (including home and building automation), and others all contribute to enabling the Internet of things. In the consumer market, IoT technology is most synonymous with products pertaining to the concept of the "smart home", covering devices and appliances (such as lighting fixtures, thermostats, home security systems and cameras, and other home appliances) that support one or more common ecosystems, and can be controlled via devices associated with that ecosystem, such as smart phones and smart speakers.

There are a number of serious concerns about dangers in the growth of IoT, especially in the areas of privacy and security, and consequently industry and governmental moves to begin to address these.

4.9.1 History:

The concept of a network of smart devices was discussed as early as 1982, with a modified Coca-Cola vending machine at Carnegie Mellon University becoming the first Internet-connected appliance, able to report its inventory and whether newly loaded drinks were cold or not. Mark Weiser's 1991 paper on ubiquitous computing, "The Computer of the 21st Century", as well as academic venues such as UbiComp and PerCom produced the contemporary vision of the IoT. In 1994, Reza Raji described the concept in IEEE Spectrum as "[moving] small packets of data to a large set of nodes, so as to integrate and automate everything from home appliances to entire factories' '. Between 1993 and 1997, several companies proposed solutions like Microsoft's at Work or Novell's NEST. The field gained momentum when Bill Joy envisioned device-to-device communication as a part of his "Six Webs" framework, presented at the World Economic Forum at Davos in 1999. later MIT's Auto-ID Center, in 1999, though he prefers the phrase "Internet for things". At that point, he viewed radio-frequency identification (RFID) as essential to the Internet of things, which would allow computers to manage all individual things. Defining the Internet of things as "simply the point in time when more 'things or objects' were connected to the Internet than people", Cisco Systems estimated that the IoT was "born" between 2008 and 2009, with the things/people ratio growing from 0.08 in 2003 to 1.84 in 2010.

The MOSFET (metal-oxide-semiconductor field-effect transistor, or MOS transistor), which was originally invented by Mohamed M. Atalla and Dawon Kahng at Bell Labs in 1959. The MOSFET is the basic building block of most modern electronics, including computers, smartphones, tablets and Internet services. MOSFET scaling miniaturization at a pace predicted by Dennard scaling and Moore's law has been the driving force behind technological advances in the electronics industry since the late 20th century. MOSFET scaling has been extended into the early 21st century with advances such as reducing power consumption, silicon-on-insulator (SOI) semiconductor device fabrication, and multi-core processor technology, leading up to the Internet of things, which is being driven by MOSFETs scaling down to nano-electronic levels with reducing energy consumption.

4.9.2 Applications:

The extensive set of applications for IoT devices is often divided into consumer, commercial, industrial, and infrastructure spaces.

4.9.3 Consumer applications:

A growing portion of IoT devices are created for consumer use, including connected vehicles, home automation, wearable technology, connected health, and appliances with remote monitoring capabilities.

4.9.3.1 Smart home:

IoT devices are a part of the larger concept of home automation, which can include lighting, heating and air conditioning, media and security systems. Long-term benefits could include energy savings by automatically ensuring lights and electronics are turned off. A smart home or automated home could be based on a platform or hubs that control smart devices and appliances. For instance, using Apple's HomeKit, manufacturers can have their home products and accessories controlled by an application in iOS devices such as the iPhone and the Apple Watch. This could be a dedicated app or iOS native applications such as Siri. This can be demonstrated in the case of Lenovo's Smart Home Essentials, which is a line of smart home devices that are controlled through Apple's Home app or Siri without the need for a Wi-Fi bridge. There are also dedicated smart home hubs that are offered as standalone platforms to connect different smart home products and these include the Amazon Echo, Google Home, Apple's HomePod, and Samsung's SmartThings Hub. In addition to the commercial systems, there are many non-proprietary, open source ecosystems; including Home Assistant, OpenHAB and Domoticz.

4.9.4 Elder care:

One key application of a smart home is to provide assistance for those with disabilities and elderly individuals. These home systems use assistive technology to accommodate an owner's specific disabilities. Voice control can assist users with sight and mobility limitations while alert systems can be connected directly to cochlear implants worn by hearing-impaired users. They can also be equipped with additional safety features. These features can include sensors that monitor for medical emergencies such as falls or seizures. Smart home technology applied in this way can provide users with more freedom and a higher quality of life. The term "Enterprise IoT" refers to devices used in business and corporate settings. By 2019, it is estimated that the EIoT will account for 9.1 billion devices.

4.9.5 Commercial application:

Medical and healthcare

The Internet of medical things (IoMT) is an application of the IoT for medical and health related purposes, data collection and analysis for research, and monitoring. The IoMT has been referenced as "Smart Healthcare", as the technology for creating a digitized healthcare system, connecting available medical resources and healthcare services.

IoT devices can be used to enable remote health monitoring and emergency notification systems. These health monitoring devices can range from blood pressure and heart rate monitors to advanced devices capable of monitoring specialized implants, such as pacemakers, Fitbit electronic wristbands, or advanced hearing aids. Some hospitals have begun implementing "smart beds" that can detect when they are occupied and when a patient is attempting to get up. It can also adjust itself to ensure appropriate pressure and support is applied to the patient without the manual interaction of nurses. A 2015 Goldman Sachs report indicated that healthcare IoT devices "can save the United States more than $300 billion in annual healthcare expenditures by increasing revenue and decreasing cost. Moreover, the use of mobile devices to support medical follow-up led to the creation of 'm-health', used to analyze health statistics."

Specialized sensors can also be equipped within living spaces to monitor the health and general well-being of senior citizens, while also ensuring that proper treatment is being administered and assisting people regain lost mobility via therapy as well. These sensors create a network of intelligent sensors that are able to collect, process, transfer, and analyze valuable information in different environments, such as connecting in-home monitoring devices to hospital-based systems. Other consumer devices to encourage healthy living, such as connected scales or wearable heart monitors, are also a possibility with the IoT. End-to-end health monitoring IoT platforms are also available for antenatal and chronic patients, helping one manage health vitals and recurring medication requirements.

Advances in plastic and fabric electronics fabrication methods have enabled ultra-low cost, use-and-throw IoMT sensors. These sensors, along with the required RFID electronics, can be fabricated on paper or e-textiles for wirelessly powered disposable sensing devices. Applications have been established for point-of-care medical diagnostics, where portability and low system-complexity is essential.

As of 2018 IoMT was not only being applied in the clinical laboratory industry, but also in the healthcare and health insurance industries. IoMT in the healthcare industry is now permitting doctors, patients, and others, such as guardians of patients, nurses, families, and similar, to be part of a system, where patient records are saved in a database, allowing doctors and the rest of the medical staff to have access to patient information. Moreover, IoT-based systems are patient-centered, which involves being flexible to the patient's medical conditions. IoMT in the insurance industry provides access to better and new types of dynamic information. This includes sensor-based solutions such as biosensors, wearables, connected health devices, and mobile apps to track customer behavior. This can lead to more accurate underwriting and new pricing models.
The application of the IOT in healthcare plays a fundamental role in managing chronic diseases and in disease prevention and control. Remote monitoring is made possible through the connection of powerful wireless solutions. The connectivity enables health practitioners to capture patient's data and apply complex algorithms in health data analysis.

4.9.6 Building and home automation:

IoT devices can be used to monitor and control the mechanical, electrical and electronic systems used in various types of buildings (e.g., public and private, industrial, institutions, or residential) in home automation and building automation systems. In this context, three main areas are being covered in literature.

- The integration of the Internet with building energy management systems in order to create energy efficient and IOT-driven "smart buildings".
- The possible means of real-time monitoring for reducing energy consumption and monitoring occupant behaviors.
- The integration of smart devices in the built environment and how they might know how to be used in future applications.

4.9.7 Industrial applications:

Industrial Internet of things, also known as IIoT, industrial IoT devices acquire and analyze data from connected equipment, (OT) operational technology, locations and people.

Combined with operational technology (OT) monitoring devices, IIOT helps regulate and monitor industrial systems.

4.9.8 Manufacturing:

The IoT can realize the seamless integration of various manufacturing devices equipped with sensing, identification, processing, communication, actuation, and networking capabilities. Based on such a highly integrated smart cyber-physical space, it opens the door to create whole new business and market opportunities for manufacturing. Network control and management of manufacturing equipment, asset and situation management, or manufacturing process control bring the IoT within the realm of industrial applications and smart manufacturing as well. The IoT intelligent systems enable rapid manufacturing of new products, dynamic response to product demands, and real-time optimization of manufacturing production and supply chain networks, by networking machinery, sensors and control systems together.

Digital control systems to automate process controls, operator tools and service information systems to optimize plant safety and security are within the purview of the IoT. But it also extends itself to asset management via predictive maintenance, statistical evaluation, and measurements to maximize reliability. Industrial management systems can also be integrated with smart grids, enabling real-time energy optimization. Measurements, automated controls, plant optimization, health and safety management, and other functions are provided by a large number of networked sensors.

Industrial IoT (IIoT) in manufacturing could generate so much business value that it will eventually lead to the Fourth Industrial Revolution, also referred to as Industry 4.0. The potential for growth from implementing IIoT may generate $12 trillion of global GDP by 2030.

4.9.9 Agriculture:

There are numerous IoT applications in farming such as collecting data on temperature, rainfall, humidity, wind speed, pest infestation, and soil content. This data can be used to automate farming techniques, take informed decisions to improve quality and quantity, minimize risk and waste, and reduce effort required to manage crops. For example, farmers can now monitor soil temperature and moisture from afar, and even apply IoT-acquired data to precision fertilization programs.

In August 2018, Toyota Tsusho began a partnership with Microsoft to create fish farming tools using the Microsoft Azure application suite for IoT technologies related to water management. Developed in part by researchers from Kindai University, the water pump mechanisms use artificial intelligence to count the number of fish on a conveyor belt, analyze the number of fish, and deduce the effectiveness of water flow from the data the fish provide. The specific computer programs used in the process fall under the Azure Machine Learning and the Azure IoT Hub platforms.

4.9.10 Energy management:

Significant numbers of energy-consuming devices (e.g. switches, power outlets, bulbs, televisions, etc.) already integrate Internet connectivity, which can allow them to communicate with utilities to balance power generation and energy usage[90] and optimize energy consumption as a whole. These devices allow for remote control by users, or central management via a cloud-based interface, and enable functions like scheduling (e.g., remotely powering on or off heating systems, controlling ovens, changing lighting conditions etc.). The smart grid is a utility-side IoT application; systems gather and act on energy and power-related information to improve the efficiency of the production and distribution of electricity. Using advanced metering infrastructure (AMI) Internet-connected devices, electric utilities not only collect data from end-users, but also manage distribution automation devices like transformers

4.9.11 Military applications:

4.9.11.1 Internet of Military Things:

The Internet of Military Things (IoMT) is the application of IoT technologies in the military domain for the purposes of reconnaissance, surveillance, and other combat-related objectives. It is heavily influenced by the future prospects of warfare in an urban environment and involves the use of sensors, munitions, vehicles, robots, human-wearable biometrics, and other smart technology that is relevant on the battlefield.

4.9.11.2 Internet of Battlefield Things:

The Internet of Battlefield Things (IoBT) is a project initiated and executed by the U.S. Army Research Laboratory (ARL) that focuses on the basic science related to IoT that enhance the capabilities of Army soldiers. In 2017, ARL launched the Internet of Battlefield Things Collaborative Research Alliance (IoBT-CRA), establishing a working collaboration between industry, university, and Army researchers to advance the theoretical foundations of IoT technologies and their applications to Army operations.

4.9.12 Architecture:

IOT System architecture, in its simplistic view, consists of three tiers: Tier 1: Devices, Tier 2: the Edge Gateway, and Tier 3: the Cloud. Devices include networked things, such as the sensors and actuators found in IIoT equipment, particularly those that use protocols such as Modbus, Bluetooth, Zigbee, or proprietary protocols, to connect to an Edge Gateway. The Edge Gateway consists of sensor data aggregation systems called Edge Gateways that provide functionality, such as pre-processing of the data, securing connectivity to cloud, using systems such as WebSockets, the event hub, and, even in some cases, edge analytics or fog computing. Edge Gateway layer is also required to give a common view of the devices to the upper layers to facilitate easier management. The final tier includes the cloud application built for IIoT using the microservices architecture, which are usually polyglot and inherently secure in nature using HTTPS/OAuth. It includes various database systems that store sensor data, such as time series databases or asset stores using backend data storage systems (e.g. Cassandra, PostgreSQL). The cloud tier in most cloud-based IoT systems features an event queuing and messaging system that handles communication that transpires in all tiers. Some experts classified the three-tiers in the IIoT system as edge, platform, and enterprise and these are connected by proximity network, access network, and service network, respectively.

Building on the Internet of things, the web of things is architecture for the application layer of the Internet of things looking at the convergence of data from IoT devices into Web applications to create innovative use-cases. In order to program and control the flow of information in the Internet of things, a predicted architectural direction is being called BPM

Everywhere which is a blending of traditional process management with process mining and special capabilities to automate the control of large numbers of coordinated devices.

4.10 Description of IR sensors:

The applications and advantages of infrared sensors are plenty; mostly these devices are utilized for various types of security systems by implementing proximity detection themes. Other important applications are for **counting objects,** or **counting revolutions** of a rotating object. In any concept, the proximity detection package contains two devices, namely infrared light emitting diode (IR LED) and infrared light/signal detector (IR sensor). The IR LED is always ON, meaning that this device is constantly emitting light and the sensor is detecting this light. The sensors can be interfaced with a trigger circuit to generate logic high/low pulses depending on the interruptions created by any object. This design of the circuit is suitable for many applications. However this design is more power consuming and is not optimized for high ranges, in this design, range can be from 1 to 10 cm, depending on the ambient light conditions. The following are the few applications:

4.10.1 Proximity sensor:

Generally proximity sensors are used for counting the objects or for counting the revolutions of a low speed running motor. These sensor packages are readily available in the market; each package contains IR LED and IR sensor. Different sensors are available to suit the requirement, the main criteria is range, as the range increases power consumption of the device also increases. Battery operated sensors range is very less, often less than 10 centi-meters. This type of sensor can be used for constructing the digital tacho meters. This sensor can be used for most indoor applications where no important ambient light is present. For simplicity, this sensor doesn't provide ambient light immunity. However, this sensor can be used to measure the speed of an object moving at a very high speed, like in industry or in tachometers. In such applications, ambient light ignoring sensors, which rely on sending 40 KHz pulsed signals cannot be used because there are time gaps between the pulses where the sensor is 'blind'.

4.10.2 Object detection using IR light:

The solution proposed doesn't contain any special components, like photo-diodes, phototransistors, or IR receiver ICs, only a couple if IR leds, an Op amp, a transistor and a couple of resistors. In need, as the title says, a standard IR led is used for the purpose of detection. Due to that fact, the circuit is extremely simple, and any novice electronics hobbyist can easily understand and build it. To prove the concept practically, as shown in the figure, we required 2 simple IR LEDs, and they must be arranged side by side to pick-up the reflected IR light. **For detecting the reflected IR light, a very simple technique is used by using another IR-LED,** to detect the IR light that was emitted from another led of the same type.

This is an electrical property of Light Emitting Diodes (LED's), which is the fact that a led produces a voltage difference across its leads when it is subjected to light. As per the following figure, the infrared signal delivered from one LED hits the object and it is reflected, another infrared LED detects the reflected signal. The signal strength depends on the current that is passing through the infrared LED; signal strength can be defined as radiating power, which is measured in mill watts. The range depends on the signal strength.

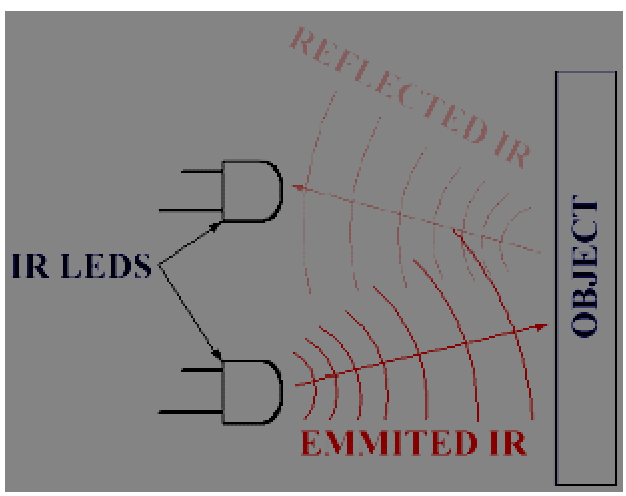

Fig 10.1 IR sensor

The infrared signal detector circuit is designed with Op-Amp (operational Amplifier), which can detect very small voltage changes also accurately.

Two different designs are proposed, each one of them is more suitable for different applications. The main difference between the 2 designs is the way infra-red (IR) light is

sent on the object. The receiver part of the circuit is exactly the same in both designs. Both the circuit's sender and receiver can be constructed on the same board. In another concept, the receiver can be kept parallel to the transmitter. The IR led's encapsulated to protect them from ambient light, this kind of encapsulation was totally sufficient to overcome all noise due to ambient light for indoor applications.

4.10.3 Wheel Encoder:

The wheel encoder can be used in many applications, it can be used as a revolutions counter, running speed can be displayed, and distance traveled also can be displayed. To achieve this, the wheel must be arranged with block and white stripes. This is a simple wheel encoder based on the idea that white stripes will reflect IR light, while black ones will absorb it, this will result in a series of electrical pulses as the wheel is rotating, providing the micro controller with precious information that can be used to calculate displacement, velocity or even acceleration. It is now clear that this kind of sensor has to be Always ON, to detect every single white stripe passing in front of it, to achieve accurate results.

4.10.4 Contact-Less tachometer:

This is a tachometer that counts the revolutions per minute of a rotating object, given that the object has a reflective stripe glued on it that will pass in front of the IR sensor for each and every revolution, giving a pulse per revolution. Again a microcontroller will have to be used to 'understand' the data provided by the sensor and display it. Many commercial contact less tachometers are designed with infrared sensors because of their cost effectiveness.

4.10.5 Contact less obstacle detection:

In this design, which is oriented to obstacle detection in robots, our primary target is to reach high ranges, from 25 to 35 cm, depending on ambient light conditions. Increasing the current flowing in the led extends the range of the sensor. This is a delicate task, as we need to send pulses of IR instead of constant IR emission. The duty cycle of the pulses turning the LED ON and OFF have to be calculated with precision, so that the average current flowing into the LED never exceeds the LED's maximum DC current (or 10mA as a standard safe value).

4.10.6 Basics of IR transmitter:

Infrared transmitter emits IR rays in planar wave front manner. Even though infrared rays spread in all directions, it propagates along a straight line in a forward direction. IR rays have the characteristics of producing secondary wavelets when it collides with any obstacles in its path. This property of IR is discussed here.

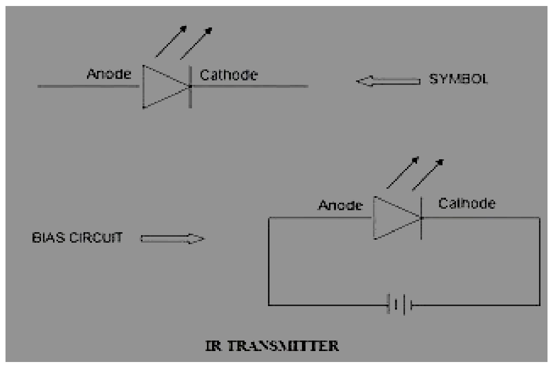

Fig 10.2 IR Transmitter

When IR rays get emitted from LEDs, they move in the direction they are angled. When any obstacle interferes in the path, the IR rays get cut and it produces secondary wavelets which propagate mostly in return direction or in a direction opposite to that of the primary waves, which produces the net result like reflection of IR rays.

4.10.7 Basics of IR receiver:

Infrared photo receiver is a two terminal PN junction device, which operates in a reverse bias. It has a small transparent window, which allows light to strike the PN junction. A photodiode is a type of photo detector capable of converting light into either current or voltage, depending upon the mode of operation. Most photodiodes will look similar to a light emitting diode. They will have two leads, or wires, coming from the bottom. The shorter end of the two is the cathode, while the longer end is the anode. A photodiode consists of PN junction or PIN structure. When a photon of sufficient energy strikes the diode, it excites an electron thereby creating a mobile electron and a positively charged electron hole. If the absorption occurs in the junction's depletion region, or one diffusion length away from it, these carriers are swept from the junction by the built-in field of the depletion region. Thus holes move toward the anode, and electrons toward the cathode, and a photocurrent is produced.

4.10.8 Use of Infrared Detectors Basics:

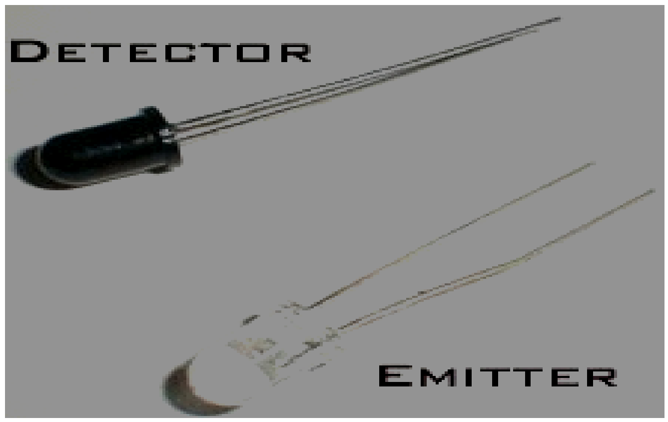

Fig 10.3 IR Transmitter and Receiver

4.10.9 IR emitter and IR phototransistor:

An infrared emitter is an LED made from gallium arsenide, which emits near-infrared energy at about 880nm.

The infrared phototransistor acts as a transistor with the base voltage determined by the amount of light hitting the transistor. Hence it acts as a variable current source. Greater amount of IR light causes greater current to flow through the collector-emitter leads. As shown in the diagram below, the phototransistor is wired in a similar configuration to the voltage divider. The variable current traveling through the resistor causes a voltage drop in the pull-up resistor.

4.10.10 This voltage is measured as the output of the device:

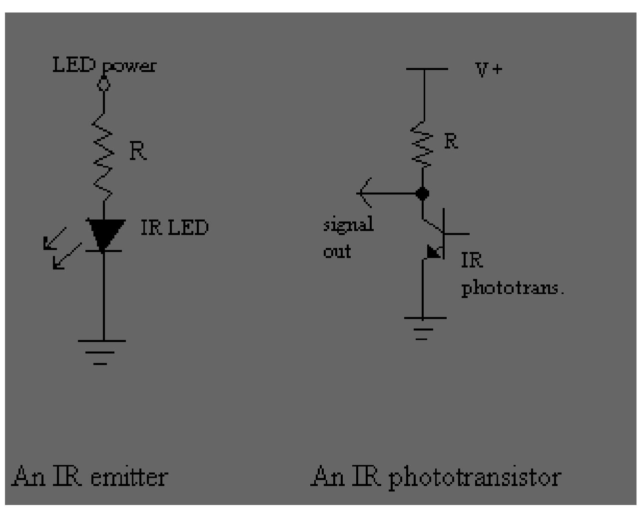

Fig 4.10.1 An IR Emitter and IR Phototransistor

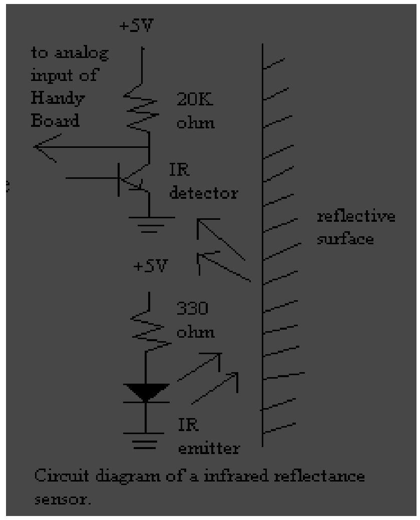

Fig 4.10.2 Circuit Diagram of reflectance sensor

IR reflectance sensors contain a matched infrared transmitter and infrared receiver pair. These devices work by measuring the amount of light that is reflected into the receiver. Because the receiver also responds to ambient light, the device works best when well shielded from ambient light, and when the distance between the sensor and the reflective surface is small (less than 5mm). IR reflectance sensors are often used to detect white and black surfaces. White surfaces generally reflect well, while black surfaces reflect poorly. One of such applications is the line follower of a robot.

4.10.11 Schematic Diagram for a Single Pair of Infrared Transmitter and Receiver:

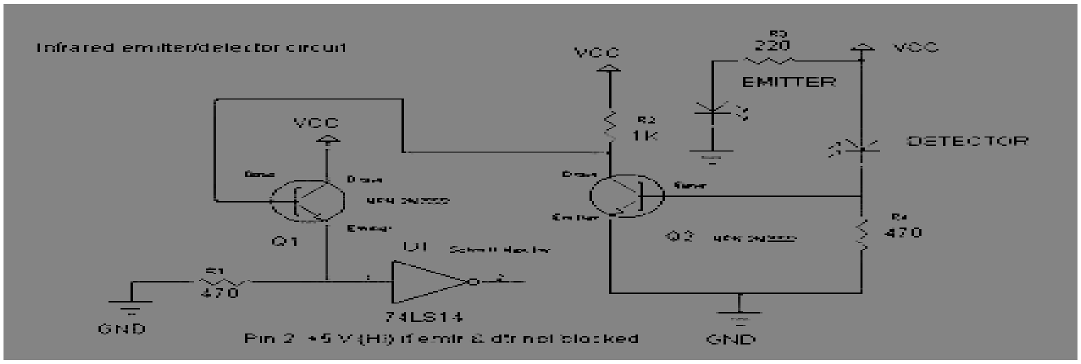

Fig 4.10.3 Infrared Emitter/Detector circuit

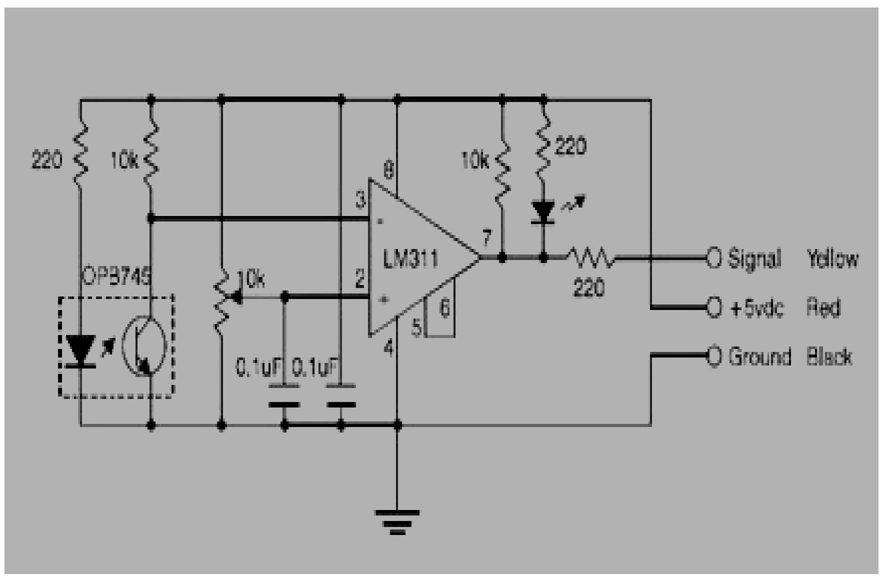

Fig 4.10.4 Operation of a typical circuit

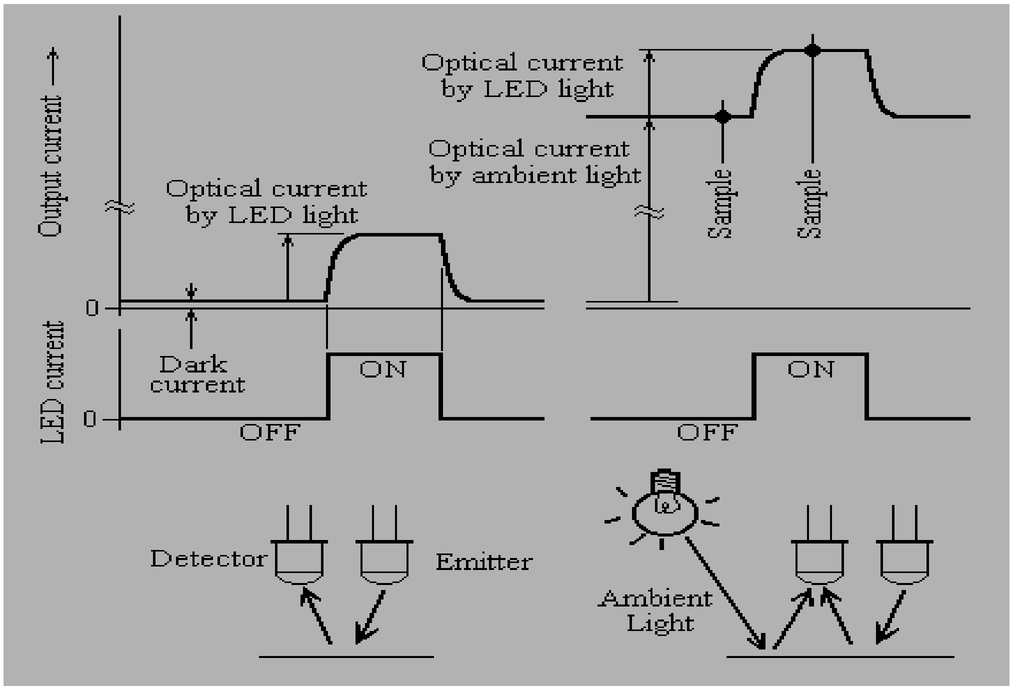

Fig 4.10.5 Optical Reflectors

If the emitter and detector (phototransistor) are not blocked, then the output on pin 2 of the 74LS14 will be high (apx. 5 Volts). When they are blocked, then the output will be low (apx. 0 Volts). The 74LS14 is a Schmitt trigger hex inverter.

4.10.12 Remote control application:

A remote control is an electronic device used for the remote operation of a machine. The term remote control can be also referred to as "remote" or "controller" when abbreviated. It is known by many other names as well, such as clicker. Commonly, remote controls are Consumer IR devices used to issue commands from a distance to televisions or other consumer electronics such as stereo systems and DVD players. Remote controls for these devices are usually small wireless handheld objects with an array of buttons for adjusting various settings such as television channel, track number, and volume. In fact, for the majority of modern devices with this kind of control, the remote contains all the function controls while the controlled device itself only has a handful of essential primary controls. Most of these remotes communicate to their respective devices via infrared (IR) signals and a few via radio signals. They are usually powered by small AAA or AA size batteries.

4.10.13 Infrared Radiation:

Infrared radiation is electromagnetic radiation whose wavelength is longer than that of visible light, but shorter than that of terahertz radiation and microwaves. The name means "below red" (from the Latin *infra*, "below"), red being the color of visible light with the longest wavelength. Infrared radiation has wavelengths between about 750 nm and 1 mm, spanning three orders of magnitude. Humans at normal body temperature can radiate at a wavelength of 10 micrometers. Infrared imaging is used extensively for both military and civilian purposes. Military applications include target acquisition, surveillance, night vision, and homing and tracking. Non-military uses include thermal efficiency analysis, remote temperature sensing, short-ranged wireless communication, spectroscopy, and weather forecasting. Infrared astronomy uses sensor-equipped telescopes to penetrate dusty regions of space, such as molecular clouds; detect cool objects such as planets, and to view highly red-shifted objects from the early days of the universe.

At the atomic level, infrared energy elicits vibration modes in a molecule through a change in the dipole moment, making it a useful frequency range for study of these energy states. Infrared spectroscopy examines absorption and transmission of photons in the infrared energy range, based on their frequency and intensity.

Objects generally emit infrared radiation across a spectrum of wavelengths, but only a specific region of the spectrum is of interest because sensors are usually designed only to collect radiation within a specific bandwidth. As a result, the infrared band is often subdivided into smaller sections.

The boundary between visible and infrared light is not precisely defined. The human eye is markedly less sensitive to light above 700 nm wavelength, so shorter frequencies make insignificant contributions to scenes illuminated by common light sources. But particularly intense light (e.g., from lasers, or from bright daylight with the visible light removed by colored gels) can be detected up to approximately 780 nm, and will be perceived as red light.

The onset of infrared is defined (according to different standards) at various values typically between 700 to 800 nm.

Infrared radiation is popularly known as "heat" or sometimes "heat radiation", since many people attribute all radiant heating to infrared light and/or to all infrared radiation to being a result of heating. This is a widespread misconception, since light and electromagnetic waves of any frequency will heat surfaces that absorb them. Infrared light from the Sun only accounts for 49% of the heating of the Earth, with the rest being caused by visible light that is absorbed then re-radiated at longer wavelengths. Visible light or ultraviolet-emitting lasers can char paper and incandescently hot objects emit visible radiation. It is true that objects at room temperature will emit radiation mostly concentrated in the 8 to 12 micrometer band, but this is not distinct from the emission of visible light by incandescent objects and ultraviolet by even hotter objects.

4.11 Description of LDR

An electronic instrumentation system consists of a number of components, which together perform a measurement and record the result. The input device receives the quantity under measurement and delivers a proportionate electrical signal. “A sensor is an electronic device which converts one form of energy (information) into another form without getting into actual physical contact with it”. An electro optical sensor is an electronic component that responds in some way or another to either light or the electromagnetic waves in the infrared ultraviolet and x-ray bands close to the visible light band. Photosensitive elements are versatile tools for detecting (radiant energy) light. They exceed the sensitivity of the human eye to all the colors of the spectrum and operate even into the ultraviolet and infrared regions. “Photoconductive cells”, also known as ‘photo resistors’ or “Light dependent resistors” is one among the photosensitive devices that is mostly used in Engineering applications. Photoconductive cells are elements halide or a metallic whose conductivity is a sulfide (e.g., cadmium sulfide, cds). When these materials are illuminated, free electrons are produced as photon energy drives them from a valence band into a conduction band.

Like in a conductor, due to free electrons current flow occurs if an electrical function of the incident electromagnetic radiation. The circuit symbol for photoresistors is a normal resistor symbol enclosed within a circle and given the Greek letter lambda (l) to denote that it responds to light. The active region of a photoconductive cell is a thin film of silicon, germanium, selenium, a metallic potential is applied. Because increased illumination creates additional free electrons, the resistance of the photoconductive cell decreases with increases in illumination level.

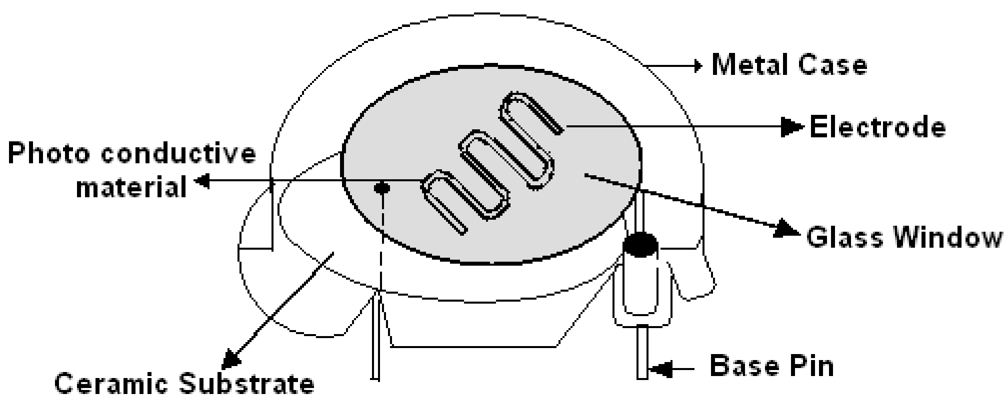

Fig 4.11.1 Photoconductive Light Detectors

The specification for a photo resistor is given as light /dark ratio. In almost all LDR'S the resistance changes from millions of ohms when dark, to hundreds of ohms under maximum illumination. The following factors must be considered while selecting photoconductive cells for a particular application.

4.11.1 SPECTRAL RESPONSE:

The spectral response of any photo sensor relates its sensitivity to light to the wavelength of electromagnetic radiation impinging on it. In most cases, the response peaks at a certain wavelength. This point is taken as unity and all other wavelengths are compared to it. Normalized means designating the peak response wavelength as either 1 or 100 percent.

For sources of multiple wavelengths (eg:- light source) a wider spectral response is required. The desirable spectral response is one that has a peak at the wavelengths of interest, and poor response at the wavelengths of ambient light.

4.11.2 SENSITIVITY:

Sensitivity is the relationship between light intensity impinging on the sensitive area of a photocell and the signal output of the cell in the circuit. The sensitivity of a photo resistor is normally measured with respect to a particular circuit or test. The definition of sensitivity can be given as "ratio of change of resistance to unit change of light illumination level.

$$S = \frac{\Delta R}{\Delta L} \; ; \qquad \Delta R = \text{Change in resistance} \qquad \Delta L = \text{Change of light level.}$$

The actual resistance is a function of the material used and the geometry of the photoresistor. The sensitivity is largely a result of the material selection.

4.11.3 RELATIVE SENSITIVITY:

The relative sensitivity of a photoconductive cell is defined as the ratio of conductance (G), of the cell at the desired measurement wavelength (G✦) to the conductance at the peak response wavelength (G✦☗) under a constant illumination level.

Sr = G✦ / G✦☗ ;

Photoconductive cells exhibit a kind of "hysteresis effect" in which the resistance change that will be caused by a given level of illumination depends on the light levels applied in the cell's past. I.e., "The present or instantaneous conductance of a cell at a specific light level is a function of the cell's previous exposure and duration of that exposure".

4.11.4 TEMPERATURE COEFFICIENT:

The temperature coefficient (▶) of any resistor is the change in resistance per unit change in temperature, usually specified in degree Celsius.

$$\tau = \frac{\Delta R}{\Delta {}^{0}C}$$

In LDR'S, the temperature coefficient is a function of both light level and the material used. As the light level changes, the temperature coefficient tends to vary inversely. Using a photoresistor in as high a light level as possible is one method of dealing with this characteristic.

4.11.5 APPLICATIONS OF LDR'S:

Light Meter Circuits , Light Operated Audio switch

Optical choppers for low Drift Amplifiers

Light Operated tone generators , Light controlled Relays.

4.12 Description of temperature sensor:

In general, a temperature sensor is a device which is designed specifically to measure the hotness or coldness of an object. LM35 is a precision IC temperature sensor with its output proportional to the temperature (in °C). With LM35, the temperature can be measured more accurately than with a thermistor. It also possesses low self heating and does not cause more than 0.1 °C temperature rise in still air. The operating temperature range is from -55°C to 150°C. The LM35's low output impedance, linear output, and precise inherent calibration make interfacing to readout or control circuitry especially easy. It has its applications on power supplies, battery management, appliances, etc.

The LM35 is an integrated circuit sensor that can be used to measure temperature with an electrical output proportional to the temperature (in °C). It can measure temperature more accurately. The sensor circuitry is sealed and not subject to oxidation. The LM35 generates a higher output voltage than thermocouples and may not require that the output voltage be amplified. The LM35 has an output voltage that is proportional to the Celsius temperature. The scale factor is .01V/°C.

The LM35 does not require any external calibration or trimming and maintains an accuracy of +/-0.4°C at room temperature and +/-0.8°C over a range of 0°C to +100°C.Another important characteristic of the LM35 is that it draws only 60 micro amps from its supply and possesses a low self-heating capability. The LM35 comes in many different packages such as TO-92 plastic transistor-like package, T0-46 metal can transistor-like package, 8-lead surface mount SO-8 small outline package.

4.12.1 Working principle of LM35:

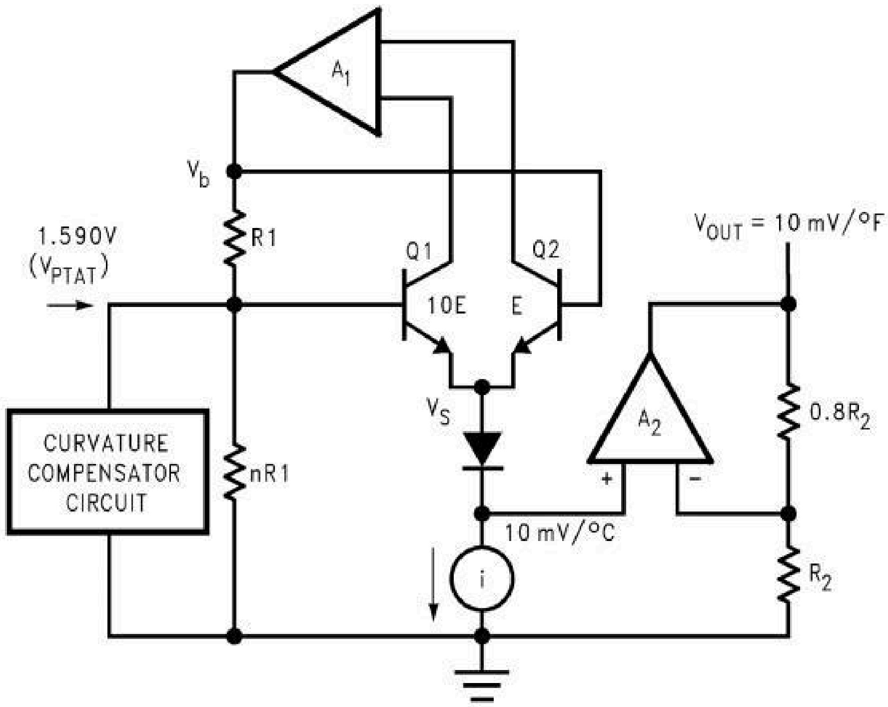

Fig 4.11.2 lM35 circuit diagram

There are two transistors in the center of the drawing. One has ten times the emitter area of the other. This means it has one tenth of the current density, since the same current is going through both transistors. This causes a voltage across the resistor R1 that is proportional to the absolute temperature, and is almost linear across the range. The "almost" part is taken care of by a special circuit that straightens out the slightly curved graph of voltage versus temperature. The amplifier at the top ensures that the voltage at the base of the left transistor (Q1) is proportional to absolute temperature (PTAT) by comparing the output of the two transistors. The amplifier at the right converts absolute temperature (measured in Kelvin) into either Fahrenheit or Celsius, depending on the part (LM34 or LM35).The little circle with the "i" in it is a constant current source circuit. The two resistors are calibrated in the factory to produce a highly accurate temperature sensor. The integrated circuit has many transistors in it -- two in the middle, some in each amplifier, some in the constant current source, and some in the curvature compensation circuit. All of that is fit into the tiny package with three leads.

The LM35 IC has 3 pins, out of 2 for the power supply and one for the analog output. It is a low voltage IC which uses approximately +5VDC of power. The output pin provides an analog voltage output that is linearly proportional to the Celsius (centigrade) temperature. Pin 2 gives an output of 1 milli-volt per 0.1°C (10mV per degree). So to get the degree value in Celsius, all that must be done is to take the voltage output and divide it by 10-this gives out the value degrees in Celsius.

Chapter – 5

Software Implementation

5.1 Arduino Integrated Development Environment (IDE)

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Hardware to upload programs and communicate with them.

5.2 Writing Sketches

Programs written using Arduino Software (IDE) are called sketches. These sketches are written in the text editor and are saved with the file extension .ino. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom right hand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.

NB: Versions of the Arduino Software (IDE) prior to 1.0 saved sketches with the extension .pde. It is possible to open these files with version 1.0, you will be prompted to save the sketch with the .ino extension on save.

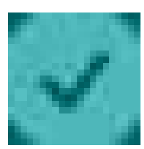

Fig 5.2.1 Verify Checks your code for errors compiling it.

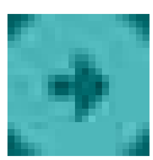

Fig 5.2.2 Upload Compiles your code and uploads it to the configured board.

Note: If you are using an external programmer with your board, you can hold down the "shift" key on your computer when using this icon. The text will change to "Upload using Programmer"

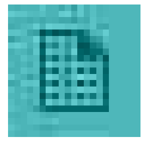

Fig 5.2.3 New Creates a new sketch.

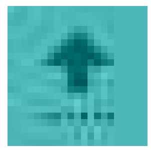

Fig 5.2.4 Open Presents a menu of all the sketches in your sketchbook.

Clicking one will open it within the current window overwriting its content.

Note: due to a bug in Java, this menu doesn't scroll; if you need to open a sketch late in the list, use the File | Sketchbook menu instead.

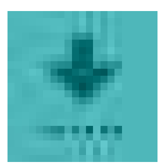

Fig 5.2.5 Save Saves your sketch.

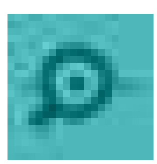

Fig 5.2.6 Serial Monitor Opens the serial monitor.

Additional commands are found within the five menus: File, Edit, Sketch, Tools, Help. The menus are context sensitive, which means only those items relevant to the work currently being carried out are available.

5.3 File:

- New Creates a new instance of the editor, with the bare minimum structure of a sketch already in place.
- Open Allows to load a sketch file browsing through the computer drives and folders.
- Open Recent Provides a short list of the most recent sketches, ready to be opened.
- Sketchbook Shows the current sketches within the sketchbook folder structure; clicking on any name opens the corresponding sketch in a new editor instance.
- Examples Any example provided by the Arduino Software (IDE) or library shows up in this menu item. All the examples are structured in a tree that allows easy access by topic or library.
- Close Closes the instance of the Arduino Software from which it is clicked.
- Save Saves the sketch with the current name. If the file hasn't been named before, a name will be provided in a "Save as.." window.
- Save as... Allows to save the current sketch with a different name.
- Page Setup It shows the Page Setup window for printing.
- Print Sends the current sketch to the printer according to the settings defined in Page Setup.
- Preferences Opens the Preferences window where some settings of the IDE may be customized, as the language of the IDE interface.
- Quit Closes all IDE windows. The same sketches open when Quit was chosen will be automatically reopened the next time you start the IDE.

5.4 Edit:

- Undo/Redo Goes back one or more steps you did while editing; when you go back, you may go forward with Redo.
- Cut Removes the selected text from the editor and places it into the clipboard.

- Copy Duplicates the selected text in the editor and places it into the clipboard.
- Copy for Forum Copies the code of your sketch to the clipboard in a form suitable for posting to the forum, complete with syntax coloring.
- Copy as HTML Copies the code of your sketch to the clipboard as HTML, suitable for embedding in web pages.
- Paste Puts the contents of the clipboard at the cursor position, in the editor.
- Select All Selects and highlight the whole content of the editor.
- Comment/Uncomment Puts or removes the // comment marker at the beginning of each selected line.
- Increase/Decrease Indent Adds or subtracts a space at the beginning of each selected line, moving the text one space on the right or eliminating a space at the beginning.
- Find Opens the Find and Replace window where you can specify text to search inside the current sketch according to several options.
- Find Next Highlights the next occurrence - if any - of the string specified as the search item in the Find window, relative to the cursor position.
- Find Previous Highlights the previous occurrence - if any - of the string specified as the search item in the Find window relative to the cursor position.

5.5 Sketch:

- Verify/Compile Checks your sketch for errors compiling it; it will report memory usage for code and variables in the console area.
- Upload Compiles and loads the binary file onto the configured board through the configured Port.
- Upload Using Programmer This will overwrite the bootloader on the board; you will need to use Tools > Burn Bootloader to restore it and be able to Upload to USB serial port again. However, it allows you to use the full capacity of the Flash memory for your sketch. Please note that this command will NOT burn the fuses. To do so a Tools -> Burn Bootloader command must be executed.
- Export Compiled Binary Saves a .hex file that may be kept as archive or sent to the board using other tools.
- Show Sketch Folder Opens the current sketch folder.

- Include Library Adds a library to your sketch by inserting #include statements at the start of your code. Additionally, from this menu item you can access the Library Manager and import new libraries from .zip files.
- Add File... Adds a supplemental file to the sketch (it will be copied from its current location). The file is saved to the
  data subfolder of the sketch, which is intended for assets such as documentation. The contents of the data folder are not compiled, so they do not become part of the sketch program.

5.6 Tools:

- This formats your code nicely: i.e. indents it so that opening and closing curly braces lineup, and that the statements inside curly braces are indented more.
- Archive Sketch Archives a copy of the current sketch in .zip format. The archive is placed in the same directory as the sketch.
- Fix Encoding & Reload Fixes possible discrepancies between the editor char map encoding and other operating systems char maps.
- Serial Monitor Opens the serial monitor window and initiates the exchange of data with any connected board on the currently selected Port. This usually resets the board, if the board supports Reset over serial port opening.
- Board Select the board that you're using.
- This menu contains all the serial devices (real or virtual) on your machine. It should automatically refresh every time you open the top-level tools menu.
- Programmer For selecting a hardware programmer when programming a board or chip and not using the onboard USB-serial connection. Normally you won't need this, but if you're burning a bootloader to a new microcontroller, you will use this.
- Burn Bootloader The items in this menu allow you to burn a bootloader onto the microcontroller on an Arduino board. This is not required for normal use of an Arduino board but is useful if you purchase a new ATmega microcontroller (which normally comes without a bootloader). Ensure that you've selected the correct board from the Boards menu before burning the bootloader on the target board. This command also sets the right fuses.

5.7 Help:

Here you find easy access to a number of documents that come with the Arduino Software (IDE). You have access to Getting Started, Reference, this guide to the IDE and other documents locally, without an internet connection. The documents are a local copy of the online ones and may link back to our online website.

5.8 Sketchbook:

The Arduino Software (IDE) uses the concept of a sketchbook: a standard place to store your programs (or sketches). The sketches in your sketchbook can be opened from the **File > Sketchbook** menu or from the **Open** button on the toolbar. The first time you run the Arduino software, it will automatically create a directory for your sketchbook. You can view or change the location of the sketchbook location from within the **Preferences** dialog.

5.9 Tabs, Multiple Files, and Compilation:

Allows you to manage sketches with more than one file (each of which appears in its own tab). These can be normal Arduino code files (no visible extension), C files (.c extension), C++ files (.cpp), or header files (.h).

Before compiling the sketch, all the normal Arduino code files of the sketch (.ino, .pde) are concatenated into a single file following the order the tabs are shown in.

5.11 Uploading:

Before uploading your sketch, you need to select the correct items from the **Tools > Board** and **Tools > Port** menus. The boards are described below. On the Mac, the serial port is probably something like **/dev/tty.usbmodem241** (for an UNO or Mega2560 or Leonardo) or **/dev/tty.usbserial-1B1** (for a Duemilanove or earlier USB board), or **/dev/tty.USA19QW1b1P1.1** (for a serial board connected with a Keyspan USB-to-Serial adapter). On Windows, it's probably **COM1** or **COM2** (for a serial board) or **COM4**, **COM5**, **COM7**, or higher (for a USB board) - to find out, you look for a USB serial device in the ports section of the Windows Device Manager. On Linux, it should be **/dev/ttyACM** , **/dev/ttyUSBx** or similar. Once you've selected the correct serial port and board, press the upload button in the toolbar or select the **Upload** item from the **Sketch** menu. Current Arduino boards will reset automatically and begin the upload. With older boards (pre-Diecimila) that lack auto-reset, you'll need to press the reset button on the board just before starting the upload. On most boards, you'll see the RX and TX LEDs blink as the sketch is uploaded. The Arduino Software (IDE) will display a message when the upload is complete, or show an error.

When you upload a sketch, you're using the Arduino **bootloader**, a small program that has been loaded onto the microcontroller on your board. It allows you to upload code without using any additional hardware. The bootloader is active for a few seconds when the board resets; then it starts whichever sketch was most recently uploaded to the microcontroller. The bootloader will blink the on-board (pin 13) LED when it starts (i.e. when the board resets).

5.12 Libraries:

Libraries provide extra functionality for use in sketches, e.g. working with hardware or manipulating data. To use a library in a sketch, select it from the **Sketch > Import Library** menu. This will insert one or more **#include** statements at the top of the sketch and compile the library with your sketch. Because libraries are uploaded to the board with your sketch, they increase the amount of space it takes up. If a sketch no longer needs a library, simply delete its **#include** statements from the top of your code.

Some libraries are included with the Arduino software. Others can be downloaded from a variety of sources or through the Library Manager. Starting with version 1.0.5 of the IDE, you can import a library from a zip file and use it in an open sketch.

5.13 Third-Party Hardware:

Support for third-party hardware can be added to the **hardware** directory of your sketchbook directory. Platforms installed there may include board definitions (which appear in the board menu), core libraries, bootloaders, and programmer definitions. To install, create the **hardware** directory, then unzip the third-party platform into its own sub-directory. (Don't use "arduino" as the sub-directory name or you'll override the built-in Arduino platform.) To uninstall, simply delete its directory.

5.14 Serial Monitor

This displays serial sent from the Arduino board over USB or serial connector. To send data to the board, enter text and click on the "send" button or press enter. Choose the baud rate from the drop-down menu that matches the rate passed to **Serial.begin** in your sketch. Note that on Windows, Mac or Linux the board will reset (it will rerun your sketch) when you connect with the serial monitor. Please note that the Serial Monitor does not process control characters; if your sketch needs a complete management of the serial communication with control characters, you can use an external terminal program and connect it to the COM port assigned to your Arduino board.

5.15 Preferences

Some preferences can be set in the preferences dialog (found under the **Arduino** menu on the Mac, or **File** on Windows and Linux). The rest can be found in the preferences file, whose location

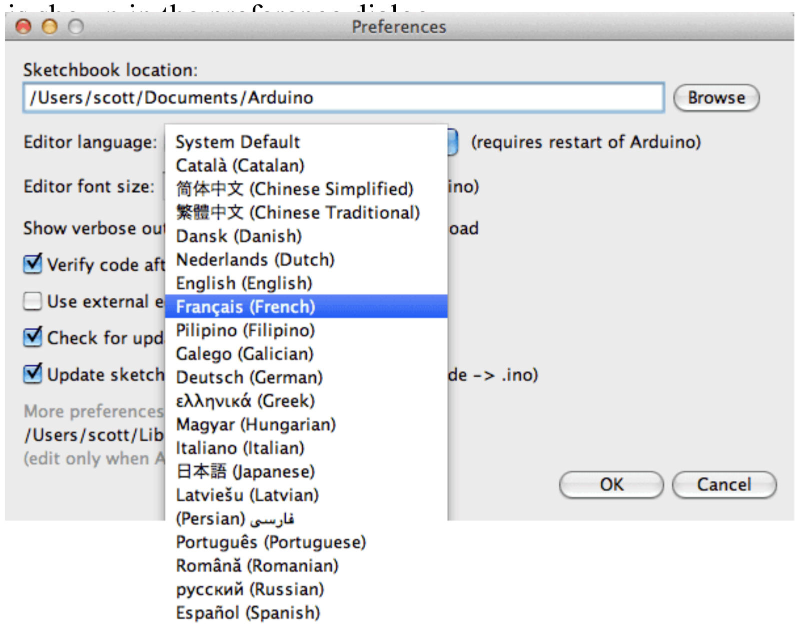

Fig 5.15.1 Language Support

Since version 1.0.1 , the Arduino Software (IDE) has been translated into 30+ different languages. By default, the IDE loads in the language selected by your operating system. (Note: on Windows and possibly Linux, this is determined by the locale setting which controls currency and date formats, not by the language the operating system is displayed in.)

If you would like to change the language manually, start the Arduino Software (IDE) and open the **Preferences** window. Next to the **Editor Language** there is a dropdown menu of currently supported languages. Select your preferred language from the menu, and restart the software to use the selected language. If your operating system language is not supported, the Arduino Software (IDE) will default to English.

You can return the software to its default setting of selecting its language based on your operating system by selecting **System Default** from the **Editor Language** drop-down. This setting will take effect when you restart the Arduino Software (IDE). Similarly, after changing your operating system's settings, you must restart the Arduino Software (IDE) to update it to the new default language.

5.16 Boards

The board selection has two effects: it sets the parameters (e.g. CPU speed and baud rate) used when compiling and uploading sketches; and sets and the file and fuse settings used by the burn bootloader command. Some of the board definitions differ only in the latter, so even if you've been uploading successfully with a particular selection you'll want to check it before burning the bootloader.

Arduino Software (IDE) includes the built in support for the boards in the following list, all based on the AVR Core. The Boards Manager included in the standard installation allows to add support for the growing number of new boards based on different cores like Arduino Due, Arduino Zero, Edison, Galileo and so on.

Chapter – 6

Results

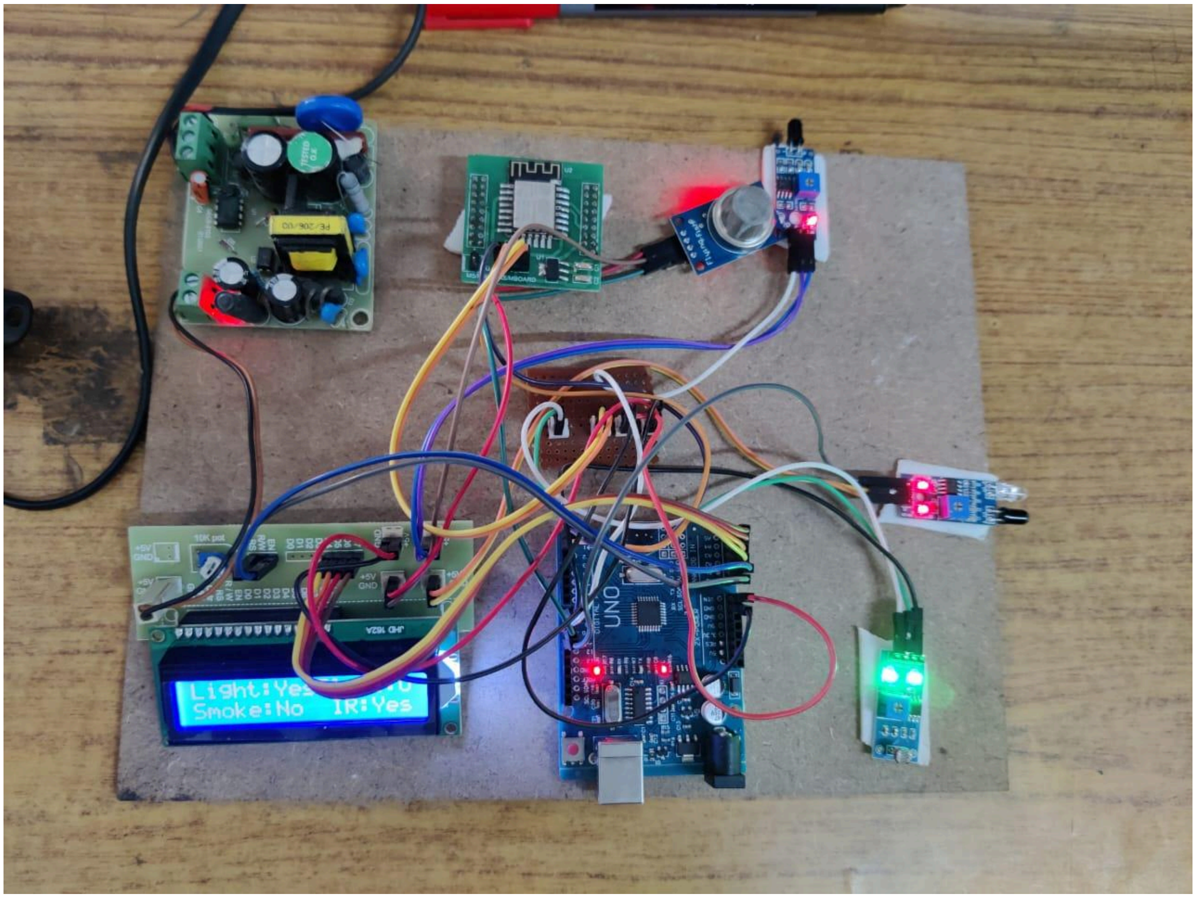

Fig 6.1 Hardware Part of the Project

In our IoT-based home security system, we have integrated an LDR (Light Dependent Resistor) to enhance night-time security. The LDR is designed to detect changes in light intensity, which makes it particularly useful for identifying unusual lighting conditions in a typically dark environment. If a thief enters the house at night and turns on any light source, such as a torch or a lamp, the LDR will immediately sense the sudden increase in light. This change is then processed by the Arduino microcontroller, which triggers an alarm and sends a notification to the homeowner via the Blynk IoT platform.

This feature ensures that even if a thief attempts to navigate in the dark by turning on a light, their presence will be quickly detected. The LDR adds an additional layer of security, making it difficult for intruders to bypass the system without triggering an alert. This proactive approach significantly enhances the overall security of the home, providing homeowners with peace of mind, especially during night-time hours.

So the LCD will show Light:YES

Fig 6.1 Light:YES,Flame:0,Smoke and IR are:NO

Our IoT-based home security system includes infrared (IR) sensors to detect unauthorized intrusions. These sensors are strategically placed near entry points to monitor for any body heat and movement. If a thief attempts to enter the house, the IR sensors detect their presence by sensing the infrared radiation emitted by their body. This detection triggers an alarm and sends an instant notification to the homeowner via the Blynk IoT platform. The IR sensors ensure real-time monitoring and immediate response to unauthorized access, significantly enhancing the security of the home and providing peace of mind to the residents.

So the LCD will show IR:YES

Fig 6.2 IR:YES,Flame:0,Smoke and Light are:NO

Our IoT-based home security system incorporates a flame sensor to detect fire hazards. This sensor continuously monitors for the presence of flames or sudden spikes in temperature. If a fire is detected, the sensor triggers an alarm and sends an immediate notification to the homeowner via the Blynk IoT platform. This allows for quick response and action, potentially preventing significant damage. The flame sensor enhances the safety of the home by providing early detection of fire, ensuring that homeowners can address the situation promptly and mitigate risks associated with fire incidents.

So the LCD will show Flame:X

Fig 6.3 Flame:X,IR,Smoke and Light are:NO

Our IoT-based home security system includes an MQ2 gas sensor to detect gas leaks. This sensor continuously monitors the air for hazardous gasses such as LPG, methane, and smoke. If a gas leak is detected, the sensor triggers an alarm and sends an immediate notification to the homeowner via the Blynk IoT platform. This prompt alert allows for quick action to prevent potential dangers like explosions or poisoning. The MQ2 sensor enhances home safety by providing early detection of gas leaks, ensuring that homeowners can address the issue swiftly and mitigate associated risks.

So the LCD will show Smoke:Yes

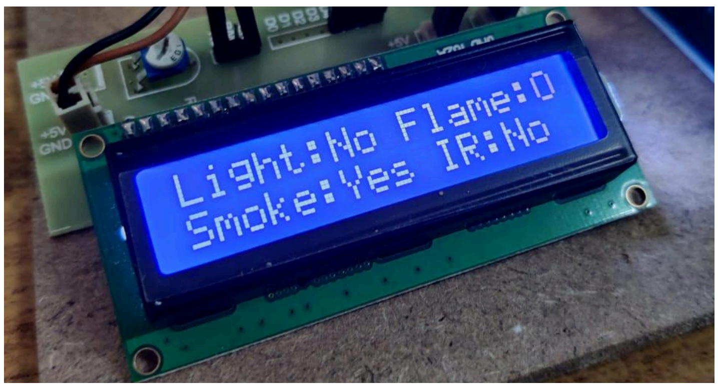

Fig 6.3 Smoke:Yes,Flame:0,IR, and Light are:NO

Chapter – 7

Conclusion and Future Scope

7.1 Conclusion:

The project work **"IoT based home security"** is completed successfully and results are found to be satisfactory. The main processing unit used in the project work is Arduino Uno board and it acts as the brain of the system and processes the data from the sensor. Also IDE (Integrated Development Environment) software is needed for Arduino based IoT projects. And we need to use ESP-8266 WiFi module to establish the WiFi communication between the Arduino and cloud platform. Arduino code is written in C++ with an addition of special methods and functions, which we'll mention later on. C++ is a human-readable programming language. When we create a 'sketch' (the name given to Arduino code files), it is processed and compiled to machine language.

IOT is a promising domain that offers exciting career options. It is profitable and has many learning opportunities. In fact, both the private and public sector companies have tremendous scope in this domain. Since digitalization is something inevitable, pursuing a career like this is highly beneficial. Most Arduino boards consist of an Atmel 8-bit AVR microcontroller (ATmega168, ATmega328, ATmega1280, or ATmega2560) with varying amounts of flash memory, pins, and features. Arduino microcontrollers are pre-programmed with a boot loader that simplifies uploading of programs to the on-chip flash memory.

IOT technology is one of the booming fields in forthcoming years and plays a major role in the field of health care & domestic security systems. IOT helps in connecting the people by empowering their health and wealth in a smart way through wearable gadgets. Recent improvements in wireless sensor networks have created a new trend in the internet of things. The main aim of this work is to provide extensive research in capturing the sensor data, analyzing the data and providing feedback to the house owners.

7.2 Future Scope:

The future scope of our IoT-based home security system includes integrating advanced AI algorithms for better threat detection and pattern recognition. We aim to incorporate more sensors for comprehensive environmental monitoring, such as humidity and CO2 levels. Enhancing the system with voice control and smart home integration will improve user experience. Additionally, developing a more robust cloud infrastructure for data analytics and expanding compatibility with other IoT devices will further enhance the system's functionality. The potential for commercialization and mass adoption in smart home ecosystems also presents a promising avenue for future development.

7.3 References:

(1) Beginning Arduino – Michael Mc Roberts

(2) – Getting started with Arduino – Mossimo Banzi

(3) The Internet of Things (IoT) is about innovative functionality and better productivity by seamlessly connecting devices. But a major threat is the lack of architecture standards for the industrial Internet and connectivity in the IoT. This article reviews recent IoT architecture evolution and what it means for industry projects.

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Page(s): 112 - 116

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Publisher: IEEE

7.4 Appendix:

```
#define BLYNK_TEMPLATE_ID "TMPL3poq1RItJ"
#define BLYNK_TEMPLATE_NAME "IOT HOME SECURE"
#define BLYNK_AUTH_TOKEN "zROVir4rj87WFu7DZSNubUmkaUJqez3V”
#include <LiquidCrystal.h>

#include <ESP8266_Lib.h>

#include <SoftwareSerial.h>

LiquidCrystal lcd( A0, A1, A2, A3, A4, A5);//rs, en, d4, d5, d6, d7

SoftwareSerial esp12(2,3); // RX, TX

#define LDR     8

#define FLAME   9

#define SMOKE   10

#define IR      11

char ssid[] = "Magni5";

char pass[] = "JHK5@magni5";

String ir,smoke,fire,ldr;

#define EspSerial Serial

#define ESP8266_BAUD 115200

BlynkTimer timer;

ESP8266 wifi(&EspSerial);

void setup()

{
```

```
pinMode(LDR, INPUT);
pinMode(FLAME, INPUT);
pinMode(SMOKE, INPUT);
pinMode(IR, INPUT);
  // Debug console
//Serial.begin(9600);
lcd.begin(16, 2);
lcd.setCursor(0, 0);
lcd.print("IOT Based       ");
lcd.setCursor(0, 1);
lcd.print("Home Security   ");
delay(3000);
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Light:Yes ");//7
lcd.setCursor(9, 0);
lcd.print("Flame:X");//15
lcd.setCursor(0, 1);
lcd.print("Smoke:Yes ");//6
lcd.setCursor(10, 1);
lcd.print("IR:Yes");//13
// Set ESP8266 baud rate
EspSerial.begin(ESP8266_BAUD);
```

```
delay(10);
Blynk.begin(BLYNK_AUTH_TOKEN, wifi, ssid, pass);
}
void myTimerEvent()
{
  Blynk.virtualWrite(ldr);
  Blynk.virtualWrite(fire);
  Blynk.virtualWrite(smoke);
  Blynk.virtualWrite(ir);
}
void loop()
{
Sensors_GetData();
Blynk.run();
myTimerEvent();
delay(2000);
}
void Sensors_GetData()
{
if(!digitalRead(LDR))
{
//Serial.println("Light Detected");
ldr="Day";
```

```
lcd.setCursor(6, 0);
lcd.print("Day");//7
}
if(digitalRead(LDR))
{
//Serial.println("No Light");
ldr="Night";
lcd.setCursor(6, 0);
lcd.print("Ngt");//7
}
if(!digitalRead(FLAME))
{
//Serial.println("Fire Detected");
fire= "Detected";
lcd.setCursor(15, 0);
lcd.print("X");//15
}
if(digitalRead(FLAME))
{
//Serial.println("No Fire");
fire= "No";
lcd.setCursor(15, 0);
lcd.print("O");//15
```

```
}
if(!digitalRead(SMOKE))
{
//Serial.println("Smoke Detected");
smoke= "Detected";
lcd.setCursor(6, 1);
lcd.print("Yes ");//6
}
if(digitalRead(SMOKE))
{//Serial.println("No Tilt Detected");
smoke= "No";
lcd.setCursor(6, 1);
lcd.print("No  ");//6}
if(!digitalRead(IR))
{//Serial.println("Person Detected");
ir= "Detected";
lcd.setCursor(13, 1);
lcd.print("Yes");//13}
if(digitalRead(IR))
{//Serial.println("No Person Detected");
ir= "Clear";
lcd.setCursor(13, 1);
lcd.print("No ");//13}
```

}